



Operate and maintain a 4-WD vehicle in safe working condition

D2.TTO.CL4.14

Trainee Manual



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The Association of Southeast Asian Nations (ASEAN) was established on 8 August 1967. The Member States of the Association are Brunei Darussalam, Cambodia, Indonesia, Lao PDR, Malaysia, Myanmar, Philippines, Singapore, Thailand and Viet Nam.

The ASEAN Secretariat is based in Jakarta, Indonesia.

General Information on ASEAN appears online at the ASEAN Website: www.asean.org.

All text is produced by William Angliss Institute of TAFE for the ASEAN Project on "Toolbox Development for Tourism Labour Divisions for Travel Agencies and Tour Operations".

This publication is supported by the Australian Government's aid program through the ASEAN-Australia Development Cooperation Program Phase II (AADCP II).

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File name: TM_Operate_maintain_a_4WD_vehicle_in_safe_wc_291015



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Introduction to trainee manual

To the Trainee

Congratulations on joining this course. This Trainee Manual is one part of a 'toolbox' which is a resource provided to trainees, trainers and assessors to help you become competent in various areas of your work.

The 'toolbox' consists of three elements:

- A Trainee Manual for you to read and study at home or in class
- A Trainer Guide with Power Point slides to help your Trainer explain the content of the training material and provide class activities to help with practice
- An Assessment Manual which provides your Assessor with oral and written questions and other assessment tasks to establish whether or not you have achieved competency.

The first thing you may notice is that this training program and the information you find in the Trainee Manual seems different to the textbooks you have used previously. This is because the method of instruction and examination is different. The method used is called Competency based training (CBT) and Competency based assessment (CBA). CBT and CBA is the training and assessment system chosen by ASEAN (Association of South-East Asian Nations) to train people to work in the tourism and hospitality industry throughout all the ASEAN member states.

What is the CBT and CBA system and why has it been adopted by ASEAN?

CBT is a way of training that concentrates on what a worker can do or is required to do at work. The aim of the training is to enable trainees to perform tasks and duties at a standard expected by employers. CBT seeks to develop the skills, knowledge and attitudes (or recognise the ones the trainee already possesses) to achieve the required competency standard. ASEAN has adopted the CBT/CBA training system as it is able to produce the type of worker that industry is looking for and this therefore increases trainee chances of obtaining employment.

CBA involves collecting evidence and making a judgement of the extent to which a worker can perform his/her duties at the required competency standard. Where a trainee can already demonstrate a degree of competency, either due to prior training or work experience, a process of 'Recognition of Prior Learning' (RPL) is available to trainees to recognise this. Please speak to your trainer about RPL if you think this applies to you.

What is a competency standard?

Competency standards are descriptions of the skills and knowledge required to perform a task or activity at the level of a required standard.

242 competency standards for the tourism and hospitality industries throughout the ASEAN region have been developed to cover all the knowledge, skills and attitudes required to work in the following occupational areas:

- Housekeeping
- Food Production
- Food and Beverage Service
- Front Office

- Travel Agencies
- Tour Operations.

All of these competency standards are available for you to look at. In fact you will find a summary of each one at the beginning of each Trainee Manual under the heading 'Unit Descriptor'. The unit descriptor describes the content of the unit you will be studying in the Trainee Manual and provides a table of contents which are divided up into 'Elements' and 'Performance Criteria'. An element is a description of one aspect of what has to be achieved in the workplace. The 'Performance Criteria' below each element details the level of performance that needs to be demonstrated to be declared competent.

There are other components of the competency standard:

- *Unit Title*: statement about what is to be done in the workplace
- *Unit Number*: unique number identifying the particular competency
- *Nominal hours*: number of classroom or practical hours usually needed to complete the competency. We call them 'nominal' hours because they can vary e.g. sometimes it will take an individual less time to complete a unit of competency because he/she has prior knowledge or work experience in that area.

The final heading you will see before you start reading the Trainee Manual is the 'Assessment Matrix'. Competency based assessment requires trainees to be assessed in at least 2 – 3 different ways, one of which must be practical. This section outlines three ways assessment can be carried out and includes work projects, written questions and oral questions. The matrix is designed to show you which performance criteria will be assessed and how they will be assessed. Your trainer and/or assessor may also use other assessment methods including 'Observation Checklist' and 'Third Party Statement'. An observation checklist is a way of recording how you perform at work and a third party statement is a statement by a supervisor or employer about the degree of competence they believe you have achieved. This can be based on observing your workplace performance, inspecting your work or gaining feedback from fellow workers.

Your trainer and/or assessor may use other methods to assess you such as:

- Journals
- Oral presentations
- Role plays
- Log books
- Group projects
- Practical demonstrations.

Remember your trainer is there to help you succeed and become competent. Please feel free to ask him or her for more explanation of what you have just read and of what is expected from you and best wishes for your future studies and future career in tourism and hospitality.

Unit descriptor

Operate and maintain a 4-WD vehicle in safe working condition

This unit deals with the skills and knowledge required to condition in a range of settings within the travel industries workplace context.

Unit Code:

D2.TTO.CL4.14

Nominal Hours:

60

Element 1: Operate a 4-wheel drive vehicle

Performance Criteria

- 1.1 Engage 4-wheel drive correctly and when appropriate
- 1.2 Drive 4-wheel drive vehicle in a variety of on-road conditions and off-road conditions
- 1.3 Provide safety and explanatory advice to passengers
- 1.4 Ensure minimal environmental impact when driving a 4-wheel drive vehicle
- 1.5 Comply with road safety and road use rules and regulations

Element 2: Recover a 4-wheel drive vehicle

Performance Criteria

- 2.1 Use a variety of techniques to recover 4-wheel drive vehicle from bogged or trapped situations
- 2.2 Optimise safety for passengers and protection of property
- 2.3 Inspect vehicle after recovery and perform necessary maintenance and repairs before proceeding

Element 3: Provide basic maintenance for a 4-wheel drive vehicle

Performance Criteria

- 3.1 Identify minor faults
- 3.2 Determine cause of faults
- 3.3 Conduct visual inspection of and operational tests on vehicle
- 3.4 Effect necessary repairs
- 3.5 Effect necessary preventative maintenance

Element 4: Report on 4-wheel drive operations

Performance Criteria

- 4.1 Complete documentation to record services and repair undertaken
- 4.2 Notify management of issues that require external attention

Assessment matrix

Showing mapping of Performance Criteria against Work Projects, Written Questions and Oral Questions

The Assessment Matrix indicates three of the most common assessment activities your Assessor may use to assess your understanding of the content of this manual and your performance - Work Projects, Written Questions and Oral Questions. It also indicates where you can find the subject content related to these assessment activities in the Trainee Manual (i.e. under which element or performance criteria). As explained in the Introduction, however, the assessors are free to choose which assessment activities are most suitable to best capture evidence of competency as they deem appropriate for individual students.

		Work Projects	Written Questions	Oral Questions
Element 1: Operate a 4-wheel drive vehicle				
1.1	Engage 4-wheel drive correctly and when appropriate	1.1	1 – 5	1
1.2	Drive 4-wheel drive vehicle in a variety of on-road conditions and off-road conditions	1.1	6 – 16	2
1.3	Provide safety and explanatory advice to passengers	1.1	17, 18	3
1.4	Ensure minimal environmental impact when driving a 4-wheel drive vehicle	1.1	19, 20	4
1.5	Comply with road safety and road use rules and regulations	1.1	21	5
Element 2: Recover a 4-wheel drive vehicle				
2.1	Use a variety of techniques to recover 4-wheel drive vehicle from bogged or trapped situations	2.1	22 – 31	6
2.2	Optimise safety for passengers and protection of property	2.1	32 – 37	7
2.3	Inspect vehicle after recovery and perform necessary maintenance and repairs before proceeding	2.1	38	8

		Work Projects	Written Questions	Oral Questions
Element 3: Provide basic maintenance for a 4-wheel drive vehicle				
3.1	Identify minor faults	3.1	39	9
3.2	Determine cause of faults	3.1	40	10
3.3	Conduct visual inspection of and operational tests on vehicle	3.1	41	11
3.4	Effect necessary repairs	3.1	42 – 46	12
3.5	Effect necessary preventative maintenance	3.1	47, 48, 49	13
Element 4: Report on 4-wheel drive operations				
4.1	Complete documentation to record services and repair undertaken	4.1	50, 51	14
4.2	Notify management of issues that require external attention	4.1	52	15

Glossary

Term	Explanation
4WD	Four-wheel drive; 4-wheel drive
AWD	All-wheel drive
A4WD	Automatic 4-wheel drive
ABS	Anti-lock braking system
DTC	Diagnostic Trouble Code
Diff	Abbreviation of 'Differential'
Duty of Care	A legally imposed obligation on the business and staff to take reasonable care to avoid causing foreseeable harm to customers/people.
Fauna	Animals and wildlife
Flora	Plants and flowers
HT lead	High Tension (spark plug) lead
kPa	Kilopascal
LSD	Limited Slip Differential
MBS	Minimum Breaking Strength
psi	Pounds per square inch
PTO	Power Take Off
OBD	On-Board Diagnostics
OEM	Original Equipment Manufacturer
Odometer	Device used to measure distance travelled by vehicle
SOP	Standard Operating Procedure
SUV	Sports Utility Vehicle

Element 1:

Operate a 4-wheel drive vehicle

1.1 Engage 4-wheel drive correctly and when appropriate

Introduction

An important aspect of driving a 4WD can be the ability to engage four-wheel drive efficiently and effectively when required.

This section discusses basic 4WD options, looks at differentials, presents reasons they are used by Tour Operators and identifies actions for putting a vehicle into four-wheel drive.



Basic options

A wide variety of vehicles are used for four-wheel drive touring – they can include:

- Diesel or petrol powered engines – the choice of fuel is very much an individual consideration with access to fuel for re-supply a primary consideration.
Personal preference is also a significant factor with many believing 'diesel is best' while many others believe the opposite.

Interesting articles contrasting relative merits of each can be found at:

http://www.outbackcrossing.com.au/FourWheelDrive/Petrol_or_Diesel_4WD.shtml - Petrol or diesel 4WD?

<http://www.4x4australia.com.au/drive/road-tests/0808/petrol-versus-diesel/> - Petrol versus diesel.

- Standard 'off-the-shelf' vehicles – which:
 - Are produced according to manufacturers' specifications
 - May be bought with a wide range of options fitted from the factory
 - Can be optioned with after-market options to suit individual need
- Vehicles produced to Tour Operator specifications – which means they are specially configured
 - For identified touring conditions
 - To accommodate the experience the Operator has built up over the years
- A range of configurations – including:
 - Small capacity vehicles (cars, SUVs, utilities, wagons, motor homes) accommodating four to five people
 - Multi-seat units (people movers, troop carriers) with space for seven to 12 people
 - Large or small coaches which can seat 20 plus.

Partial 4-wheel drive

Partial 4WD is also known as part-time 4WD.

It means the vehicle is usually a 2-wheel drive vehicle when travelling normal roads/highways but has the capacity to become a 4WD when 4WD is selected.

The two wheels which are not under power are known as free-wheeling hubs.



High and Low range

Part-time 4WDs generally give the driver the option of selecting 'High' or 'Low' range 4WD.

Low range provides/generates more torque at the wheels through lower gear ratios which is useful for driving difficult terrain at low speeds.

High range gives greater control and traction when driving quicker on surfaces such as compacted sand/beaches.

2WD

A vehicle that is only powered by one set of wheels (either front or rear) is known as a two-wheel drive/2WD.

Permanent 4-wheel drive

A permanent 4-wheel drive vehicle is normally in 4WD mode.

There is therefore no need to engage 4WD because it is always selected.

There can be the facility to de-select 4WD and engage 2WD.

Permanent 4-wheel drives are also referred to all-wheel drive (AWD) vehicles.

Automatic four-wheel drive

This is a system where the vehicle senses different individual wheel speeds and engages 4WD automatically it detects a need to do so.

These are known as A4WDs.

4x4 and 4x2

- 4x4 = a vehicle that has four wheels (the first 4) and has drive to all 4 wheels (the second 4)
- 4x2 = a conventional 2WD vehicle with 4 wheels and power only to two of them
- 6x6 = a vehicle that has six wheels and has drive to all 6 wheels.

Differentials

Understanding of the role of the differential/s in a vehicle (also known as the 'diff/s') is central to understanding the operation of a 4WD.

Background

The engine develops power which is transferred to a drive shaft. The drive shaft is connected to the axles using a differential which transfers the power in the drive shaft to axles, and on to the wheels.

The gears in the differential allow the drive wheels on opposite sides of the vehicle to rotate at different speeds to accommodate optimum performance when turning corners or in slippery conditions.



4WD vehicles have (at least) two diffs – one for the front wheels and one for the rear wheels. Some makes and models also have a third diff (a centre diff) which enables the front and rear drive shafts to also turn at different speeds, as well as letting each wheel rotate/drive at different speeds.

Limited Slip Diffs

Some vehicles may be fitted with LSDs.

The LSD allows more power from the drive shaft to be sent to the wheel with the *most* traction in a situation where one of the wheels is slipping or spinning.

This means extra (but limited) drive can be gained from the vehicle via the wheel which is getting the most purchase on the road/track surface.

Diff locking

Diff locking is where the differential is locked so it drives both the axles (and hence both the wheels) with the same speed/power regardless of the road surface/condition.

Depending on the make/model of the 4WD it can be possible to:

- Lock the front diff
- Lock the rear diff
- Lock the centre diff.

This facility is often a benefit in certain four-wheel driving conditions as it can give much better traction, drive, control and ability to get through sand, bogs and difficult terrain.

Some vehicles (A4WDs) are fitted with automatic diff lockers meaning the vehicle senses when they are needed, and turns them on and off as required on the basis of information provided by sensors.

It is important to unlock manually applied diff locks when the difficult driving conditions are overcome and the vehicle returns to 'normal' use (so the wheels are able to rotate at different speeds as required when turning),

Websites

More information can be found at:

<http://www.lcool.org/technical/diffs/diffs.html> - Land Cruiser Owners Online: Diffs for Dummies

<https://www.youtube.com/watch?v=gIGvhvOhLHU> – How a differential works and types of differentials

<https://www.youtube.com/watch?v=WeLm7wHvdxQ> – Understanding limited slip differentials

<https://www.youtube.com/watch?v=RcRBYpccP6g> – How diff locks work and when to use it

Reasons for using 4WDs

Tourist Operators choose to use 4WDs for a variety of reasons including:

- Their customers expect them to be using one
- Customers like the often unique nature of an off-road 4WD experience, driving through water and mud
- Four-wheel drives give customers confidence in the vehicle and the fact the tour will be a safe and trouble-free one
- They give greater ground clearance and ramp-over angles than conventional vehicles
- They allow better approach and departure angles when (for example) approaching river crossing and clearing jump-ups
- They enable more dependable manoeuvrability of off-road, difficult and challenging conditions such as loose sand and mud – meaning a 4WD can take people places a conventional 2WD vehicle cannot go.



Context for four-wheel driving

It is necessary to understand:

- Four-wheel drive needs to be selected **before** it is needed (see below) – based on:
 - Observation and analysis of the upcoming conditions of the route to be taken
 - Personal experience
 - Getting into trouble and then selecting 4WD is to be avoided – as:
 - It may be difficult/impossible to select 4WD when the vehicle is stuck
 - It may not resolve the situation
- Momentum is important – four-wheel drive *alone* is insufficient to get a vehicle through some conditions
- Tyre pressure is important in some situations – generally, lowering pressure in tyres will result in better traction through an increase in the footprint of the tread of the tyres on the surface
- Vehicles should be operated strictly in accordance with any specifications provided for by the manufacturer of the vehicle as stated in the Owner/User manual
- There are limits as to what a 4WD can do – they cannot ‘do anything and go anywhere’
- Four-wheel drive may need to be supplemented by fitting chains to the tyres in some conditions such as snow or mud – to give added traction and control.

Situations possibly requiring engagement of four-wheel drive

There can be a need to engage 4WD:

- When tracks are muddy/boggy
- If the terrain is rough and rocky
- Where there is soft sand
- In the event of snow and ice
- Going up or down steep inclines
- When crossing creeks/rivers.

See more in next section.



Engaging 4WD

The actions required for engaging four-wheel drive will vary depending on the make and model of the vehicle.

Always:

- Follow manufacturer instructions
- Create a comfortable driving/sitting position enabling full operation of all foot and hand-operated controls (knobs and pedals, levers and switches) and clear line of sight around the vehicle through windscreens, windows and mirrors
- Practice the procedure before there is a need to implement it in a real-life tour situation
- Refrain from engaging optional 4-wheel drive on dry, hard, flat and firm surfaces – as this increases fuel consumption and wear and tear on parts.

The following are indicative (only) of actions which may be required:

- Stopping the vehicle – apart from vehicles which have ‘on the fly/move’ 4WD allowing engagement of four-wheel drive while travelling below a nominated speed
- Moving gear shift to neutral
- Locking in the hubs of the vehicle – by getting out of the vehicle and moving the locking hubs from the ‘Free’ position to the ‘Lock’ position
- Moving 4WD selector to either High range or Low range (or Lock), or
- Pushing a designated 4WD button on the dash, or
- Moving a selector knob to the required position
- Selecting appropriate gear.

Always ensure the safety of passengers and vehicle is protected/maintained when engaging 4-wheel drive.

1.2 Drive 4-wheel drive vehicle in a variety of on-road conditions and off-road conditions

Introduction

Drivers of four-wheel drive tour vehicles need to be proficient in driving in a variety of on- road and off-road situations.

This section provides information about competent driving in a range of possible conditions

Golden keys

The Golden Keys to successful four-wheel driving are the two related factors of:

- Practice
- Experience.

It is **critical** drivers of 4WD tour vehicles gain competency in driving their vehicle in all expected on- and off-road conditions *before* they do so in a real-life touring environment where paying passengers/tour groups are involved.



Off-road and on-road

Off-road conditions refer to terrain/ground that is not sealed or paved.

On-road conditions refer to driving on sealed roads/highways.

On-road conditions

Driving 4WDs in on-road conditions may include driving:

- In dry conditions – where arguably most driving is done, and the standard approach is to:
 - Use two-wheel drive/H2 only
 - Leave extra distance between vehicles to allow for the extra braking distances required for heavier 4WD vehicles
 - Take tight corners at a lower speed due to the relatively high centre of gravity for 4WDs
 - Drive with headlights on at all times
- In wet conditions such as when there is rain – where speed must be reduced and High range 4WD can be used
- In snow and ice – where speed must be reduced, chains may need to be fitted and Low range may need to be selected.

Off-road conditions

Operating a 4WD in off-road conditions can require driving:

- Steep inclines – including ascending and descending
- Sand
- Rocky terrain
- Water crossings
- Very soft ground
- Mud.



Pre-requisites

Before driving off-road it is advisable to (where possible/practicable):

- Ensure the vehicle has been properly prepared for the conditions which are known to exist or which are expected on the trip – this may include:
 - Loading the vehicle with necessary recovery gear
 - Fitting protection and bash plates to the underside of the vehicle
 - Fitting appropriate tyres
 - Having tyre deflators and pumps
 - Having warning/sand flags
 - Having chains
 - Fitting a snorkel
 - Fitting a viscous fan
 - Fitting extended differential breathers
- Undertake basic checks – to confirm:
 - The vehicle is fully operational – to the nest extent possible under the circumstances that apply at the time
 - Safety and recovery gear is available and operational
- Inspect or consider the way ahead – to:
 - Determine the route/path to be taken
 - Identify potential obstacles/problems
 - Make the final 'Go' or 'No go' decision
- Always be prepared to, as required:
 - Choose or find an alternate route
 - Back-track and go through using another track
 - Wait for poor conditions (high water levels, high flow rates of rivers, deep snow) to abate/improve before proceeding

- Consider passenger safety – which may involve:
 - Communicating with them to advise them of what to expect
 - Providing safety information
 - Keeping them involved and included in the tour experience
 - Taking **controlled** risks can be part of the excitement and experience of a 4WD adventure but they must never be undertaken where they pose an uncontrolled threat to people/passengers
- Engage four-wheel drive – as appropriate for the individual vehicle and the nature of the route and terrain to be driven.



Driving inclines

Ascending steep inclines

Indicative actions are:

- Inspect the path – looking for:
 - A suitable route through/across the track which presents itself – taking into account, for example, ruts, holes, fallen trees, rocks
 - Surface condition – in relation to whether the ground is wet or dry and whether it is hard underfoot or soft
 - Approach angle – to ensure the vehicle can begin the climb and get its wheels onto the incline, and not get 'beached'/stuck trying to make the climb
 - General angle/severity of the climb
 - Length of the incline – in terms of the distance to be covered to complete the climb
 - Possible recovery points that can be used – if the vehicle becomes stuck
- Engage 4WD – which may require locking in hubs
- Engage diff locks – if fitted and if necessary
- Turn off air conditioner – to avoid inconsistent engine acceleration
- Begin the climb in a manner that reflects the findings of observation of the incline – which may require, as appropriate:
 - Reversing and taking a run at the incline to gain momentum perhaps in H4 where the ground is solid, the length of the climb is short and the angle of incline is not too steep
 - Selecting Low range first or second gear and driving slowly up the incline where it is a steep incline, ground is soft sand/or the distance to be covered is significant
- Stay square to the incline – climbing at an angle to the rise introduces the potential for the vehicle to roll
- Work the wheel a little from side-to-side if the wheels start spinning – in order to use the tread on the sides of the tyres to gain additional purchase/traction

- Pause at the top of the incline – in order to:
 - Check the descent – if there is one (see below)
 - Check for oncoming vehicles.



Stall recovery

If the vehicle stalls when climbing a steep incline the procedure for a safe stall recovery is:

- Apply foot brake as vehicle stalls
- Apply handbrake
- Turn ignition off
- Depress clutch
- Select reverse gear
- Release clutch
- Release handbrake slowly
- Turn on ignition and release brakes
- Start vehicle allowing it to reverse down the slope
- Apply brakes as necessary to control descent.

Descending steep inclines

Standard advice regarding steep descents is to avoid them where possible and find an alternate route especially where the route is wet or otherwise presents an extra risk on top of the angle of descent.

Indicative actions may include:

- Inspect the path – looking for:
 - A suitable route through/across the track which presents itself – taking into account, for example, ruts, holes, fallen trees, rocks
 - Surface condition – in relation to whether the ground is wet or dry and whether it is hard underfoot or soft
 - Departure angle at the end of the incline – to ensure the vehicle can leave the incline and make it to level ground instead of being stuck half-way between the incline and the flat ground
 - Length of the descent – in terms of the distance to be covered to complete the descent
 - General angle/severity of the descent
 - Possible run-off points (into banks, bushes, trees) – in case the descent becomes out-of-control
- Engage 4WD – which may require locking in hubs
- Engage diff lock
- Turn off air conditioner – to avoid inconsistent engine acceleration

- Begin travel in a manner that reflects the findings of observation of the descent – which may include:
 - Selecting Low range first gear
 - Allowing vehicle to creep forward under its own power without using the accelerator – letting the engine act as a brake
 - Applying extra braking as required using either foot brake or hand brake – to keep a safe speed
 - Minimising use of brake – allowing engine to do the braking
 - Avoiding locking of wheels – the wheels must continue to rotate in order to retain control/steering
 - Keeping wheels pointing forward
 - Stay square on to the incline – to avoid potential for roll-over.



Stall recovery

If the vehicle stalls when descending a steep incline the procedure for a safe stall recovery is:

- Turn ignition off
- Apply foot brake
- Engage handbrake
- Depress clutch
- Select low gear
- Release clutch
- Slowly release handbrake
- Turn ignition on
- Release brakes
- Start vehicle and allow to continue down the slope
- Apply brakes emulating ABS as necessary to control descent
- Steer into skids.

Websites

See more information at:

<http://www.4x4australia.com.au/gear/1004/4x4-steep-slopes-tips/> - 4x4 steep slope tips

<http://4xoverland.com/how-do-4x4s-work/4x4-driving/> - How to drive a 4x4 off road [Steep slopes]

<https://www.youtube.com/watch?v=uTW3H4WBxnY> – 4WD Hill Climb Comparo

<http://www.iventureclub.com.au/4x4-tips/article.html?k=Jem4UGra2DY> – 4x4: Tip- Steep terrain

Driving on sand

Preparing for sand

When preparing for soft sand:

- Lower tyre pressure of 4WD vehicle – as a starting point to increase the grip area of the tread and help keep it on top of the sand instead of sinking (or biting) into it:
 - Reduce by half standard on-road pressure
 - Reduce more for softer surfaces
- Lower tyre pressure of trailer, where appropriate
- Use pressure gauge to determine pressures – so there is a known reference point: write the pressures down as they are obtained and trialled
- Be prepared to lower pressures further if first decrease appears insufficient – reduce by approx. two to four psi (15 – 30 kPa) each time.



Read more at:

<http://www.offroadaussie.com/2013/02/4wd-tips-and-tricks-tyre-pressure/> - 4WD tricks and tips: tyre pressure.

Driving on hard sand

Driving on compacted sand is similar to driving on sealed roads in many ways.

Techniques for driving a 4WD on hard sand include:

- Reduce tyre pressure slightly – to increase surface area of tread that makes contact with the sand
- Engage 4WD – which may require locking in hubs
- Select High range
- Drive off choosing gear/s appropriate to conditions, weight of vehicle, route being used and desired speed of travel
- Travel at a reduced speed – that is, slower than on the open highway
- Avoid contact with salt water wherever possible – to prevent potential of vehicle being damaged by rust: always wash any vehicle as soon as possible that has been contacted by salt water
- Re-inflate tyres when sand driving is completed – if they have been tyre pressure was initially reduced.

Driving on soft sand

Indicative actions/techniques are:

- Deflate tyres – to around 16 psi/110kPa or lower depending on sand depth/condition
- Attach sand/warning flags to the vehicle if driving in areas where there are hills/dunes – to help indicate location of the vehicle to others in the area
- Check recovery gear is available – to use should the vehicle become stuck
- Engage 4WD – which may require locking in hubs
- Select Low range – second gear is a good starting point
- Start slowly – to avoid wheel spin and digging in
- Drive slowly but maintain momentum
- Avoid sudden turns
- Drive in established tracks – where the sand will be more compacted by the vehicles which have previously used that route
- Be prepared for less feedback from the steering wheel – due to the depth of the sand
- Approach any dunes head on/square on if possible – to avoid potential for roll-over
- Stay alert for oncoming traffic over the top of dunes
- Never assume the drop off on the other side of a dune is the same as the climb up the dune – drop-offs on the other side can be substantially different due to wind action which may have created a sheer fall/drop away on the other side
- Use the brake gently – to avoid sand building up in front of the tyres
- Be prepared to reverse back over the tracks made by the vehicle if stuck – as these will be more compacted and may provide sufficient traction to develop the required traction and momentum to move forward
- Re-inflate tyres when sand driving is completed – if they have been tyre pressure was initially reduced.



Websites

See more information at:

<http://4xoverland.com/how-do-4x4s-work/4x4-driving/> - How to drive a 4x4 off road [Sand driving]

<https://www.youtube.com/watch?v=iT0Zph4h-L4> – ExpeditionOz 4WD Skills: Sand driving

<https://www.youtube.com/watch?v=I2bA6aC4g50> – Soft sand driving

https://www.youtube.com/watch?v=KHI_cMrvsrM – 4x4: How to drive on sand tracks

http://www.outbackcrossing.com.au/FourWheelDrive/How_To_Drive_On_Sand.shtml - How to drive on sand

https://www.youtube.com/watch?v=Int_D9zKjiY – Tips for beginners: Driving on sand, dunes and beaches

<http://www.iventureclub.com.au/4x4-tips/article.html?k=YqKAOI33YuQ> – 4x4 tip: Sand driving.

Driving across rocky terrain

In general, rocky terrain should be avoided wherever possible as it poses lots of risks in terms of:

- Uncomfortable ride for passengers
- Slow travel times – which may negatively impact itinerary
- Punctures
- Damage to the vehicle especially the chassis and sub-structure/under-side parts of the vehicle such as diffs and fuel tanks.



When driving rocky terrain:

- Inspect the path – as described above
- Reduce pressures in tyres
- Engage 4WD – which may require locking in hubs
- Engage diff locks – if fitted and if necessary
- Turn off air conditioner – to avoid inconsistent engine acceleration
- Begin in a manner that reflects the findings of observation – which may require, as appropriate:
 - Selecting Low range first or second gear
 - Driving slowly ('crawling') and smoothly to avoid wheel spin
 - Intelligently choosing the path to take to avoid traps, rocks and beaching of the vehicle
- Be prepared to ask passengers to get out of the vehicle – if patches of the route are dangerous
- Be prepared to ask passengers to act as spotters – to assist with providing directions on where to position wheels
- Try to stay on the high parts – straddling any ruts which may be present
- Re-inflate tyres when the rocky terrain has been negotiated.

Websites

Check out the following:

<http://www.4-wheeling-in-western-australia.com/rock-crawling.html> - Rock crawling and rocky obstacles (including viewing of all videos linked to this site)

<http://www.iventureclub.com.au/4x4-tips/article.html?k=1ImWwpAqbd0> – 4x4 tip: Rocks and ruts

<http://www.tough toys.com.au/how-to-4x4/driving-rocky-terrain/> - Driving rocky terrain.

Doing water crossings

Preparing for water

Standard requirements before making deep-water crossings are;

- Know the vehicle and how deep water the water can be before the vehicle begins to float
- Know how fast the water is flowing – to determine safety and work out speed required and angle of attack
- Know the surface to be travelled – on the bottom of the river and on exit from the crossing
- Know where the air intake is – to make sure the vehicle does not ingest water
- Check-out the crossing – walk through the crossing to determine depth, identify obstacles and work out the path to be taken.

Walk across the river where the left-hand wheel will go.

Walk back where the right-hand wheel will go.

- If in doubt, avoid the crossing.

Standard cautions

Be prepared to avoid the crossing if:

- The current is too fast
- The water is too deep
- The approach or departure angles are too steep
- The approach or departure are too soft or otherwise too difficult
- The vehicle is not/cannot be adequately prepared.

Driving techniques

When driving through water there can be a need to:

- Let the engine cool down first – the steps below will assist with this
- Tie a tarpaulin across the front of the vehicle – to help create a bow wave
- Loosen the fan belt – to prevent damage to the radiator (unless vehicle is fitted with a viscous fan)
- Spray the motor with water repellent (WD-40) to help preserve the electrics
- Attach recovery straps to the vehicle – in case there is a need to recover the vehicle from the middle of the river: make sure these straps are secured above the water line
- Lower windows – as a safety factor
- Advise passengers not to wear seat belts
- Lower pressure in tyres – if exit is sandy/soft
- Select 4WD, low range – first or second gear depending on type and size of engine
- Lock the diffs



- Enter the water at a slow to moderate speed – so as to create a bow wave but not a ‘splash’ which will throw water up and over the bonnet/vehicle
- Drive slowly and evenly
- Avoid using the clutch – as water will enter the housing and cause damage, slippage
- Dry brakes after exiting the water.



Website

Take the time to visit:

<http://www.iventureclub.com.au/4x4-tips/article.html?k=zFA1eN-qEzg> – 4x4 Tip: Water crossing

<http://www.tough toys.com.au/how-to-4x4/water-crossings/> - Water crossings

<https://www.youtube.com/watch?v=wNtJNuhlf4I> – ARB Water crossing techniques

<http://www.outbacktravelsaustralia.com.au/driving-towing-4wd-driving-skills/water-crossings> - Water crossings

<http://www.4-wheeling-in-western-australia.com/river-crossings.html> - River crossings [including videos on this page]

<http://www.4wdonline.com/A/Water.html> - Water crossing

<http://www.pps.net.au/4wdencounter/4wdtech/water.html> - 4WD Encounter Australia: River crossings.

Driving on very soft ground or through mud

Techniques/indicative action includes:

- Inspecting the route – looking for:
 - Ruts and obstacles
 - Condition of the bottom under the surface mud/depth of the mud
 - Distance to be covered
 - Recovery points
- Dropping tyre pressure – to around 24 psi/165kPa
- Engaging 4WD
- Locking diffs
- Selecting Low range – first or second gear
- Entering the mud at a slow to moderate speed – avoiding excessive speed
- Using or not using ruts as appropriate – sometimes it is best to drive in the ruts and sometimes it is best to avoid them, for example:
 - Driving in ruts (which can provide a more compacted surface and therefore increase traction) if there is sufficient clearance for the bottom of the vehicle
 - Straddling the ruts – where the *adjacent* ground is firmer and will give better traction

- Maintaining momentum – by applying acceleration
- Spinning tyres – where mud tyres are fitted to help clean the tread
- Being prepare to stop, reverse and re-try a route if progress is halted – as this may give extra traction through the more compacted bottom surface
- Moving the steering wheel left-to-right – to allow side lugs of tyres to bite the mud.

Website

See more information at:

<http://4xoverland.com/how-do-4x4s-work/4x4-driving/> - How to drive a 4x4 off road [Mud driving]

https://www.youtube.com/watch?v=O7iqHO8_R8 – Mud driving the right and wrong way

<http://www.iventureclub.com.au/4x4-tips/article.html?k=hmcpHxSX7pc> – 4x4 Tip: mud driving

<http://www.toughtoys.com.au/how-to-4x4/mud-driving-techniques/> - Mud driving techniques



1.3 Provide safety and explanatory advice to passengers

Introduction

Drivers of tour vehicles are obliged to protect their own safety, the safety of all passengers and the safety of others when operating their tour vehicle.

This section discusses Duty of Care requirements, provides hints on how to advise passengers about vehicle and 4WD safety and presents generic safety tips.



Duty of Care

Common law imposes an obligation on all vehicle drivers to:

- Create and maintain a touring environment that does not pose a risk to people (staff, members of the public, customers)
- Take action to avoid causing foreseeable harm to people/customers/tour group members or their property/belongings while they are on tour
- Avoid engaging in risky behaviour or actions which will pose a threat to anyone.

In practice this means the driver needs to/has responsibility for ensuring:

- The vehicle is safe/roadworthy when the tour departs
- The safety/roadworthiness of the vehicle is maintained through the trip
- All road rules and speed limits are obeyed
- The vehicle is driven in a safe and responsible manner
- Obvious risks are avoided or controlled
- Passengers are informed about potential risks/hazards
- Passengers comply with safety rules
- Cargo is stored/stowed appropriately – which may require:
 - Re-packing/re-locating items for off-road 4-wheel drive travel to attain a lower centre of gravity for the vehicle
 - Placing items behind physical protection barriers – or otherwise securing them to prevent injuring people or causing vehicle damage.

Consequences of failing to meet Duty of Care obligations

If drivers of 4WDs fail to meet their Duty of Care obligations:

- They may be sued by passengers/those they injure or who lose property for negligence
- The Tour Operator they work may be sued for not discharging their Due Diligence obligations by properly training their drivers.

The outcomes of successful civil claims for negligence are commonly the awarding of money ('damages') to the person who has been injured or suffered loss.

Advising passengers about vehicle safety

Vehicle safety can/should be communicated to tour group members:

- Verbally – in terms of:
 - Being part of the standard tour briefing prior to departure of the tour from the Tour Operator's office
 - Being mentioned at the daily tour briefing (held every morning before the tour heads off for the day) on the first day there is expectation four-wheel drive will need to be engaged
- In written form – by:
 - Providing a handout sheet
 - Including vehicle/passenger safety as a heading in the standard tour advice/information provided to all passengers when they buy their tickets
 - Detailing safety requirements on the business website
- By screening a video – showing:
 - Aspects of vehicle/passenger safety
 - Expected touring conditions
 - How 4WDs perform in certain conditions.



Safety information

Safety details which might be communicated to passengers can include advising them:

- To use/wear seat belts at all times – except when river/creek crossings are being undertaken
- To stay completely inside the vehicle at all times when the vehicle is moving – with specific reference to ensuring arms, hands and heads are kept inside the vehicle
- To keep windows wound up when travelling through bush – to prevent branches flicking people in the head as they brush against trees and bushes
- To wind windows down when undertaking river/creek crossings – to facilitate easier egress from the vehicle if there is a problem
- To use hand straps/holds when the vehicle is traversing uneven ground
- To follow the driver's instructions/directions at all times – fully and promptly
- To refrain from altering any vehicle settings – or tampering with/touching any of the controls of the vehicle
- To secure loose personal items when the vehicle is in motion – advising how and where this may be done
- To refrain from throwing items/litter from the vehicle
- To refrain from behaviour that would be distracting to the driver of the vehicle
- To only get in and out of the vehicle when it has come to a full stop
- About on-board emergency items, first aid kits, recovery equipment, communication systems and emergency exits
- About Emergency Management Plans which have been developed – to guide action in the event of an emergency or accident.

1.4 Ensure minimal environmental impact when driving a 4-wheel drive vehicle

Introduction

In an era where Tour Operators are focussed on sustainable and responsible tourism, it is important drivers of 4WD vehicles ensure minimal environmental impact when on tour.

This section identifies generic impacts which may be caused by 4WD vehicles/tours and presents techniques to minimise the negative impact of 4WDs.



Possible generic impacts

It is possible 4WDs may impact the environment in the following ways:

- Disturbing local communities – and other visitors in the area
- Damaging/destroying flora – which can also lead to erosion of the land
- Causing noise pollution
- Contaminating fresh water sources
- Harming soil structure – by compacting, dislodging it and/or causing pot-holes
- Scaring, injuring or killing local wildlife.

Techniques to minimise negative impact

Not all impacts can be avoided or eliminated but with planning, care and consideration most can be minimised or mitigated.

The following techniques have proved to assist with environmental harm minimisation:

- Adhering to designated tracks and routes in sensitive areas, parks or other prescribed areas – as opposed to creating new tracks and driving off-road
- Assessing quality of tracks or intended routes before travelling on them – and refraining from travelling on tracks which are:
 - Closed
 - Degraded or boggy
- Minimising the effects of noise – by:
 - Ensuring exhaust and muffler system is effective
 - Avoiding revving/racing engines
 - Staying clear of other users
- Ensuring all required permits (including payment of fees), authorities and advice have been obtained – and their requirements have been understood and complied with
- Abiding by local prohibitions – regarding, for example:
 - Entry by vehicles into nominated areas
 - Use of tracks under certain weather conditions

- Being vigilant when preparing, planning and driving – so as to avoid causing damage to:
 - Property and structures
 - Natural habitats of animals
 - Natural assets
 - Ensuring proper behaviour of passengers – in relation to applicable Codes of conduct which may require:
 - They do not litter the route and areas travelled
 - They do not remove flora and/or fauna from the area
 - They do not use firearms
 - Being patient – and:
 - Not speeding
 - Using passing bays, where provided, rather than driving off-road and creating new tracks in order to get passed other vehicles
 - Taking the time to close gates which had to be opened – standard protocol is to leave all gates as they were found
 - Removing obstructions on tracks – as opposed to driving round them and creating new tracks
 - Not using trees as winch points if a recovery is required – or using tree protectors if no other suitable point is available
 - Giving way to foot traffic on tracks – and allowing animals to have right of way
 - Driving on beaches only on the firm sand below high water marks – so high tide will remove vehicle marks.
- Note:** while driving below the high tide mark will reduce 4WD impact many people recommend this is not done in case the vehicle becomes stuck/breaks down in which case it may be swamped with water when the tide comes in.
- Adhering to weight limits for vehicles and trailers – to avoid extra depression of ruts/soil
 - Taking care when driving through dry grass – that the vehicle's exhaust system does not start fires
 - Not travelling at night – when there is greater danger of hitting wildlife.



1.5 Ensure minimal environmental impact when driving a 4-wheel drive vehicle

Introduction

All drivers of 4WDs must ensure they comply with the traffic laws as imposed by the country they are operating in.

This section identifies basic requirements in this regard.

Compliance requirements

Drivers of four-wheel drive vehicles are required to:

- Be properly licensed – according to:
 - The size of the vehicle
 - Type/class of vehicle
 - Whether it is manual or automatic
- Carry their licence – and present it on demand to authorities
- Ensure the vehicle is correctly registered – in accordance with:
 - The type/class of vehicle being used
 - The commercial nature of the vehicle
- Display proof of registration – to demonstrate current and relevant registration
- Ensure the vehicle complies with roadworthy requirements – as these apply to:
 - Vehicle safety
 - Passenger safety
- Obtain, display and comply with applicable special permits as/if required by local laws – for use of designated:
 - Parks
 - Tracks
 - Local roads
 - Beaches
 - Other routes/locations
- Comply with all applicable road rules – as these apply to:
 - Adhering to speed limits
 - Obeying traffic signs
 - Giving right of way to pedestrians and other vehicles – and passing and over-taking
 - Using indicators, lights and warning and safety devices
 - Demonstrating consideration and tolerance for other road users



- Comply with applicable fatigue management regulations – so the driver has sufficient:
 - Rest/sleep
 - Breaks from driving
- Comply with applicable environmental protection legislation – as this applies to:
 - Smoke emissions
 - Noise pollution
- Comply with relevant drink-driving legislation – and does not drive contrary to applicable blood alcohol limits as these apply to:
 - Age of the driver
 - Licence status
 - Type/class of vehicle being driven
- Comply with legislation prohibiting driving while under the influence of drugs.



Work Projects

It is a requirement of this Unit you complete Work Projects as advised by your Trainer. You must submit documentation, suitable evidence or other relevant proof of completion of the project to your Trainer by the agreed date.

- 1.1 To fulfil the requirements of this Work Project you are required to operate a 4-wheel drive vehicle approved by your Trainer/Assessor providing evidence you have:
- Engaged 4-wheel drive in various conditions correctly and when appropriate
 - Driven the vehicle in a designated range of on- and off-road conditions effectively and safely
 - Provided safety briefings and explanatory advice to 4-wheel drive passengers as part of vehicle operation
 - Ensured minimal environmental impact when driving the vehicle
 - Complied with applicable road safety and road use rules while driving the vehicle.
-

Summary

Operate a 4-wheel drive vehicle

When operating a 4-wheel drive vehicle:

- Take time to learn about and understand the vehicle
- Locate and read the User/Operator manual for the vehicle
- Know how to engage and disengage four-wheel drive
- Be aware of the ranges available once 4WD has been selected
- Understand diff locking options available on the vehicle
- Select 4WD before it is needed
- Realise 4WD vehicles can become stuck/trapped
- Drive to avoid problems not overcome them
- Choose correct tyre pressure
- Ensure vehicle is properly prepared and equipped with necessary recovery gear
- Obtain practice and experience before going on tour
- Check the vehicle and check the route before proceeding
- Make passenger safety and comfort a primary consideration
- Adapt driving to suit the conditions
- Minimise environmental impact
- Observe all road safety and road use rules

Element 2: Recover a 4-wheel drive vehicle

2.1 Use a variety of techniques to recover 4-wheel drive vehicle from bogged or trapped situations

Introduction

It is a fact of life when four-wheel driving that there can be a need to recover the vehicle when it becomes bogged or trapped.

This section presents information and advice to assist with vehicle recovery from a range of common problem 4WD situations.



Golden keys

The Golden Keys to successful recovery of four-wheel drive vehicles are the two factors identified earlier:

- Practice
- Experience.

It is **critical** drivers of 4WD tour vehicles gain competency in vehicle recovery in all expected on- and off-road conditions *before* they do so in a real-life touring environment where paying passengers/tour groups are involved.

Recovery situations

There can be a need to recover vehicles from situations where they have become bogged, beached or otherwise trapped:

- In sand
- In mud/soft ground
- In water
- On rocky surfaces.

Recovery gear knowledge

The vehicle driver must:

- Be aware of the gear available on/in the vehicle for recovery situations
- Know where the recovery gear is located
- Know the recovery points/hooks and mounting features/points of the vehicle – which are connected directly to the vehicle chassis to give optimum strength
- Prepare/attach any gear which is deemed necessary/possibly required for an upcoming situation

- Check the gear for safety and readiness for use – especially:
 - Checking for frayed cables and straps
 - Verifying winches which have been immersed in water on previous tours have been checked and serviced
 - Making sure items which may be required are readily accessible.



Ratings

It is important to use recovery gear that is rated correctly for the vehicle and the recovery situation.

If incorrect/insufficiently rated gear (cables, straps, mounting points, winches, jacks) is used there is danger posed through:

- The snapping/breaking of items
- Items being ripped off vehicles
- Jacks collapsing under weight
- Winches failing under load.

Rating refers to the weight (in tons or tonnes) of the vehicle being recovered known as Minimum Breaking Strength (MBS).

Manufacturers may also rate their products in generic classifications such as:

- Standard
- Heavy Duty
- Extra heavy duty.

Examples of recovery gear

Vehicles can be equipped with a range of recovery gear.

Common practice is to match the recovery fitted to/stowed in the vehicle with the recovery situations which are likely to be encountered along the way.

For example, a tour which does not feature sand driving would not load sand ladders or sand mats onto the vehicle.

Recovery gear may include:

- A shovel – traditionally a basic long-handled shovel
- Tyre deflators – to lower tyre pressures before attempting to drive out of a bogged/trapped situation
- Rated shackles ('D' shackles or 'Bow' shackles) appropriately rated to the weight to be handled – used to attach recovery cables/straps to vehicle mounting points and to attach snatch straps to tree protectors. Never use these to connect two snatch straps together.

- Drag chains – used to tow another vehicle or to remove (for example, fallen trees) from the track
- Snatch straps – which are used in combination with another vehicle to extricate ('snatch') a trapped vehicle free from sand or mud
- Winches and cables – including electric winches, PTO winches, hydraulic winches, capstan ('wheel') winches, hand-operated winches and snatch blocks
- Inflatable bags – inflated by pumps or the exhaust
- Jacks – including standard scissor and bottle jacks as well as high-lift jacks (also known as 'kangaroo jacks')
- Recovery tracks and/or sand ladders – for sand and bog recoveries as well as for bridging gaps
- Accessories – such as:
 - Winch gloves – to protect hands when handling winch cables
 - Winch extensions – to give extra length to winch cables
 - Extension straps – to give extra length to snatch straps
 - Snatch strap joiners – to assist in the joining of snatch straps so they can be easily disconnected
 - Tree protectors – to wrap around trees to protect them when winching out
 - Equaliser straps – effectively allows snatch straps to be connected to two recovery points on the vehicle being recovered instead of just one thereby spreading the load
 - Recovery blankets/dampers (also known as a cable dampener) – placed over cables and snatch straps when winching or snatching to absorb energy and protect by-standers in the event a winch cable or strap breaks when under load
 - Ground anchors – used to help recover a vehicle where there are no natural recovery points/trees to attach a winch cable to.



See:

<https://www.youtube.com/watch?v=CDqu3La1nls> – 4WD Recovery series Part 1: The essential gear

Using long-handled shovel

The long-handled shovel is a basic recovery tool often used as a first-response when vehicles are stuck, or to even out tracks to prevent vehicles *becoming* trapped.

They are used for:

- Digging away sand or dirt – from under the wheels of vehicles
- Getting rid of debris from tracks/routes
- Removing unwanted soil or material from approach and departure angles of river crossings and steep inclines.

Websites

Take time to view:

<http://www.iventureclub.com.au/4x4-tips/article.html?k=cSwbJ7Nu3UQ> 4x4: Solo recovery.

Using snatch straps

Using a snatch strap can involve:

- Attaching strap to vehicle mounting/recovery points only on both vehicles – and not to tow ball, bull bar or bumper bar
- Unrolling the strap between the two vehicles making sure the strap is straight, not twisted and clear between the vehicles – and is not kinked or blocked by obstacles
- Adding another snatch strap – where extra length is required between the recovery vehicle and the trapped/bogged vehicle
- Attaching strap to vehicles:
 - Using a rated shackle – as opposed to simply looping over an attachment point
 - Securing to central mounting/recovery points on both vehicles
- Using long-handled shovel to remove build up in front of the wheels – in the direction the recovery will occur
- Position the recovery vehicle so there is about three metres of slack in the snatch strap – to allow for a run-up to 'snatch' the trapped vehicle from its position
- Placing recovery damper/blanket across snatch strap in the middle third part of the strap – for safety reasons
- Ensuring effective communication between the recovery vehicle and the trapped vehicle – so the two drivers can coordinate their actions
- Starting the trapped/vehicle and engaging 4WD, Low range, first gear
- Confirming between drivers their readiness to start the procedure
- Driving the recovery vehicle forward at a medium/moderate rate in 4WD, Low range, second gear – as opposed to racing forward quickly at high speed
- Accelerating the trapped vehicle so it can assist the recovery vehicle when the snatch strap takes effect
- Realising the vehicle does not have to be recovered/freed on the first attempt – it is possible/may be necessary to repeat and repeat to free the trapped vehicle.



Websites

See details at:

<https://www.youtube.com/watch?v=-lre31FrEAI> – Snatch strap recovery technique

<https://www.youtube.com/watch?v=bRRhfz4tdRg> – Ask an expert 031: Attaching a snatch strap

<https://www.youtube.com/watch?v=oT6R6yTg6R0> – Mad Matt's Snatch Strap Recovery Tips

<https://www.youtube.com/watch?v=i7GSL3o1pUU> – 4WDRecovery series Part 2: Snatch strap recovery technique

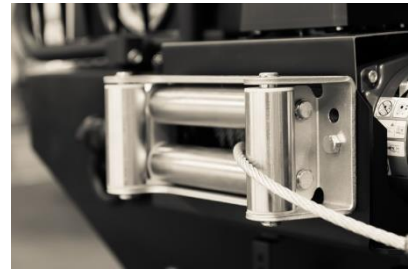
Using winches

The electric winch

The electric winch is an option for most 4WDs and where fitted is located at the front of the vehicle.

This type of winch operates independently of the vehicle engine.

The winch allows the vehicle to recover itself and can also be used for recovering other vehicles. When using the winch to recover another vehicle, the winch vehicle should be secured to an anchor point to prevent it moving/slipping.



Winch cables (or ropes) are usually attached to trees in order to winch vehicles from trapped positions but may be secured to any firm anchor point.

When using the winch for a self-recovery indicative considerations and actions are:

- Always wear gloves
- Always follow manufacturer's instructions – especially as these apply to:
 - Safety issues
 - Operation of the winch
 - Use of attachments
- Unwind the cable off the winch – giving sufficient length to attach to the recovery point: add extra lengths of cable if necessary
- Place tree protector around the tree – and attach the winch cable with a rated shackle
- Drape a winch dampener over the cable – as a safety precaution
- Take up the slack on the cable between the tree and the trapped vehicle – guide the cable as it winds back on to the winch to ensure it rolls on evenly and smoothly
- Verify passengers are out of danger
- Start the recovery by engaging the winch – opinions vary and there are two basic options:
 - Allow the winch to do all the work and recover the vehicle on its own
 - Put the vehicle in 4WD Low range first or second gear and gently accelerate to assist the action of the winch
- Have a person (tour crew member) monitor the take up of the cable on the winch spool/drum – to ensure an even wind on
- Be aware of the heavy draw on electric power of the winch – so:
 - Leave engine running throughout the recovery operation
 - Minimise its use
 - Be prepared to use it in small bursts, stopping in between to run the engine and re-charge the battery
- Use a pulley block at the anchor point ('double line pull') to assist where necessary – which will halve the load on the winch: put a dampener over both cables.

Websites

View the videos at:

<https://www.youtube.com/watch?v=AKLtDFz5WgU> – 4WDRecovery series Part 3: Basic winching techniques

https://www.youtube.com/watch?v=L2l_-3lpHII – 4WDRecovery series Part 4: Advanced winching techniques

<https://www.youtube.com/watch?v=StZHIHD09XM> – 4x4 The basic guide to winching techniques

<https://www.youtube.com/watch?v=jatVOGInI64> – 4WD winching techniques with Pat Callinan – Ray's Outdoor

<https://www.youtube.com/watch?v=StZHIHD09XM> – The basic guide to winching techniques [for electric winches]

<https://www.youtube.com/watch?v=F4mXgnll3e4> – How to winch a 4x4

https://www.youtube.com/watch?v=_4tl3HtgdcY – Reverse winch

<https://www.youtube.com/watch?v=Td92wGxvujU> – Snatch block in action

<https://www.youtube.com/watch?v=JY4cKc4FKSM> – Going 4WD: The Bush Winch/Wheel Winch

<http://www.outbacktravellaustralia.com.au/driving-towing-recovery-techniques/safe-winchng> - Safe winching

<https://www.youtube.com/watch?v=4TtAgDp6VKA> – The intelligent winch: 4x4.



The PTO winch

A PTO winch is driven by the engine of the 4WD and operates the winch drum/spool through the gearbox of the vehicle via a power take-off.

PTO winches are also mounted at the front of the vehicle.

They are a good option where long, heavy winching work is expected but have one major short-coming in that they cannot be used if the engine of the vehicle is dead/will not start.

The same protocols described for the use of the electric winch apply to the use of the PTO winch.

Websites

Visit and view:

<https://www.youtube.com/watch?v=xDAzHXkk3LE> – PTO winches

<https://www.youtube.com/watch?v=GOMZQldEctg> – Red Warrior hydraulic winch with PTO power.

Hydraulic winch

Hydraulic winches are powered by a hydraulic motor/pump which is, in turn, powered by the engine of the vehicle.

This style of winch is not a common option but seen by many as one of the best options in that it:

- Allows the same control over the winch as for an electric winch
- Does not pose the same problem with flattening the battery so it can be used for extensive lengths of time.

The down-side is the engine of the 4WD needs to be operational meaning it will not work if the engine cannot be started.

Websites

See more at:

<http://www.4wdonline.com/Equip/winches.html> - 4WD Online: Winches

<http://www.4wdonline.com/Equip/winch-TJM.html> - TJM hydraulic winch.

Capstan winch

A capstan winch features a drum/spool plus a capstan around which the cable/winch rope is fed when recovering the vehicle.

The use of a capstan means vehicle recovery points can be 'at an angle' to the vehicle without this causing a problem.

They are generally powered off the engine of the vehicle and not via an electric motor.



Hand-operated winch

Some vehicles may carry a hand-operated winch but they are uncommon as:

- They are bulky to carry
- Take more time to use when recovering a vehicle than other options
- Are limited in the length of cable that is available for use in a recovery.

When using a hand-operated winch:

- Follow manufacturer's instructions
- Wear gloves
- Attach cable hook to recovery mounting on vehicle
- Run out the cable to the tree which is to be used as the recovery point
- Attach tree protector to recovery point with rated shackle
- Thread cable through the winch – setting levers in the correct position to allow the cable to be fed through the winch
- Attach the winch to the tree protector using the shackle
- Clamp the winch onto the cable – by moving the appropriate lever on the winch
- Drape cable damper over the winch cable

- Insert handle into winch
- Pump/move the handle to commence the winching process.

Note: snatch blocks/pulleys can be used in combination with hand winches in the same way they can be used with other winches.

Websites

See:

<https://www.youtube.com/watch?v=5UzX3F4Oyqw> – 4WD Action: 114 Terrain Tamer Hand Winch

<https://www.youtube.com/watch?v=k39wFI4MQQg> – ask an expert: hand winching

<https://www.youtube.com/watch?v=MvU2g4dyKhl> – Exacta hand winch.

Using the high-lift jack

For vehicle recovery trolley jacks, bottle jacks and scissor jacks are of limited or no use.

In order to recover a trapped vehicle using a jack the standard approach is to use a high-lift jack (or kangaroo jack) because this gives a much greater lift distance.

When using these jacks:

- Clean and spray with WD-40 – as these jacks are usually carried on the outside of the vehicle and become dirty
- Follow manufacturer's instructions
- Fit jack into base plate – if base is separate
- Provide solid surface on which to locate the base of the jack – as best as possible given the conditions
- Flip out the lift plate/bar on the jack – which will be used to take the weight when the vehicle is lifted
- Locate jack as required near lifting point, which may be:
 - Solid part of the vehicle such as a steel bull bar
 - A hole/lug into which the lift bar is inserted
 - Adjacent to the bogged wheel – where a lift strap ('lift mate') can be attached to the wheel, instead of to a vehicle lift point
- Move selector/lever to 'lift'
- Insert jack handle and pump to lift vehicle – taking care:
 - To lift slowly
 - To use both hands to optimise control over the handle
 - To not allow head to be positioned above the handle – which can kick back and cause injury to the operator
 - To keep the jack stable – and not allow it to topple



- Apply pressure to the end of the handle – for maximum leverage
- Listen for the 'click' of the mechanism to indicate end of one stroke and need to start next pump
- Lift vehicle until it is clear of obstruction/out of the trapped situation
- Insert rocks, wood or proprietary products (portable tracks and/or sand ladders) under wheel and in direction of travel
- Lower jack – by:
 - Shifting lever to reverse position
 - Pumping the handle slowly
 - Keeping head away from handle
 - Taking greater care as the handle/jack is not fully weighted and can provide a greater danger of causing injury should it become out-of-control during the lowering process.



Websites

See:

- <https://www.youtube.com/watch?v=DgjqhU5PgkE> – ARB hi-lift jack demo
- https://www.youtube.com/watch?v=1a_5lpDzPPI – How to use a hi-lift jack safely
- <https://www.youtube.com/watch?v=hUL-l9Yi3Ws> – How to use the high-lift jack
- <https://www.youtube.com/watch?v=9FRMK7ZstaY> – High lift jack recovery: off road recovery

Using the high-lift jack as a winch

It is also possible to use the high-lift jack as a hand winch for some recoveries.

See:

<https://www.youtube.com/watch?v=OVY0QQiN93o> – Using high-lift jack as winch to get Discovery out

<https://www.youtube.com/watch?v=bOcQHqLx90g> – Hi-lift as a winch

<https://www.youtube.com/watch?v=pAGp8lizSQU> – Basic self-recovery using a hilift

Using bags

Under-vehicle inflatable bags are available to help raise bogged/trapped vehicles, or to use an alternative to a scissor or bottle jack when changing a wheel.

Bags may be inflated:

- Using the exhaust gases of the vehicle/engine
- Using an air compressor.

They need to be used in combination with rocks, wood or proprietary products (portable tracks and/or sand ladders) under wheel/s and in direction of travel for them to be really useful: simply lifting a trapped vehicle is not of much use – it needs to be able to move forward and the rocks, wood, tracks or ladders will allow this to happen.

Indicative instructions for using a bag jack:

- Unpack the bag from its carrier
- Locate in appropriate position under the vehicle
- Check to ensure:
 - No jagged edges/protrusions which can harm or harm the bag
 - Bag is away from hot exhausts and engine parts
- Attach hose to bag
- Start vehicle engine in neutral – if using exhaust to lift the vehicle
- Take other end of hose to exhaust pipe and jam/hold the cone of this end in position over end of the exhaust pipe
- Watch as bag inflates
- Remove hose from exhaust pipe when bag is inflated
- Insert rocks, wood or proprietary products (portable tracks and/or sand ladders) under wheel and in direction of travel
- Loosen hose connected to bag – to deflate the bag slowly and lower the vehicle.



Websites

Extra detail can be found at:

<https://www.youtube.com/watch?v=aLLTIKPU3Zs> – Exhaust jack

<https://www.youtube.com/watch?v=CqBQ8q2mxuY> – X-Jack and X-Trax

<https://www.youtube.com/watch?v=Hc7Tni99dlQ> – LROTV High lift jack versus Air jack

https://www.youtube.com/watch?v=gmsMx9_TkaU – How to use an exhaust air-jack for recovery.

Using sand ladders or light-weight tracks

Sand ladders and other proprietary light-weight tracks can be:

- Placed under the wheels of stranded vehicles to give the wheels better purchase when trying to un-stick the vehicle by either towing/winch it out or driving it out
- Used to provide a bridge to allow a vehicle to travel across an otherwise insurmountable gap.

To use the ladders/tracks:

- Shovel away build up from in front of the tyre/s – if possible
- Raise the vehicle – if possible
- Position the ladders/tracks under and in front of the wheels
- Lower the vehicle onto the ladders/tracks
- Make sure vehicle is in 4WD Low range first gear
- Drive forward
- Repeat as required until vehicle is free.

Websites

Watch the use of ladders/tracks at:

<https://www.youtube.com/watch?v=NLe71YjrUkI> – Muputrax: multi-purpose recovery tracks/sand ladders

https://www.youtube.com/watch?v=kMFa_pylslo – LROTV: Sand ladders demonstration

<https://www.youtube.com/watch?v=95m6VHEWvFk> – Overland journal: sand ladder test

https://www.youtube.com/watch?v=hm_u0wb3kD0 – Terra Trax sand ladders demo



2.2 Optimise safety for passengers and protection of property

Introduction

Passenger safety and the protection of property must always be a primary consideration when recovering a trapped vehicle.

This section explains the dangers inherent in recoveries and provides safety advice to make the recovery process safer.



Recovery dangers

Recovering a trapped vehicle can present various risks/dangers depending on where and how the vehicle is stuck.

Possible dangers may include:

- Risk of the vehicle becoming flooded in a tidal river or beach situation as the water rises
- Danger of the vehicle slipping and rolling forward, backwards or sideways
- Chance of vehicle trapped in water being pushed over or swept away by the force of the water (current/flow) against its side
- Possibility of winch cable, snatch strap or tow rope breaking and springing back causing injury to people in vehicle or by-standers
- Passengers injuring themselves as they leave/get out of a stranded vehicle – and slip/slide on muddy, wet or otherwise slippery surfaces.

Danger Zone

It is useful to consider the concept of 'Danger Zone' when recovering a trapped vehicle.

The Danger Zone is the area in which passengers/people could be injured if something goes wrong with the recovery.

Standard practice is to consider the factors which impact on defining the Danger Zone for each recovery situation and then:

- Move everyone out of that zone
- Position them behind walls, rocks or other suitable protection.

Factors to be considered include:

- Type of recovery equipment used – which indicates possibility of (cable or rope or other) failure/s
- Direction of recovery to be attempted for the trapped vehicle
- Potential for vehicle to roll, slip, topple or fall – which may be related to:
 - Fall/lay of the land
 - Surface conditions
 - Physical position of the vehicle

- Effect of suction on cable/winch capacity
- Level of difficulty of the recovery.

Golden Rule

The Golden Rule when conducting a recovery or driving a four-wheel drive is 'people safety is paramount'.

This means:

- Doing nothing to jeopardise the safety of humans
- Safety of people comes before protecting their property
- Safety of people comes before the safety of animals
- Passengers must be protected before the safety of the tour vehicle
- Not allowing time pressures of a tour itinerary to impact on decisions and actions required to safely address a recovery situation
- Not allowing financial considerations to dictate an unsafe course of action is chosen over a known, more expensive but safer recovery option.

Context and preparation

Every trapped/recovery situation is different and presents its own unique challenges, problems and potential dangers.

An experienced driver will have sufficiently scouted/checked out the route ahead to identify the potential for problems and will not be surprised when they eventuate thereby enabling them to be better mentally (and otherwise) prepared to respond.

Preparation for situations where the vehicle could become trapped may include:

- Check and prepare recovery gear which may be required – to ensure it is:
 - Quickly available
 - Fit for use should it be needed
- Check out/walk or otherwise inspect the route
- Re-locate items in the vehicle – in order to:
 - Alter the balance/weight distribution
 - Secure items
 - Give ready access to certain gear
- Advise passengers of identified threats
- Give necessary safety advice – such as:
 - Wind windows down
 - Keep doors unlocked
 - Keep seat belts on or off – depending on the situation.



Important first responses

If the tour vehicle becomes trapped the driver/tour crew must:

- Take control of the situation
- Show leadership and confidence regarding the situation
- Avoid panic
- Inspect the situation to determine and gain first-hand knowledge and an understanding of the situation – rather than guess at what it might be
- Work out a plan of action
- Not rush – but take a deep breath and think things through
- Act – rather than react.



Maintaining passenger safety and protecting their property

Depending on the rescue situation, there may be a need to:

- Remove passenger belongings from the vehicle and/or trailer – and remove to a safer position
- Instruct passengers to leave the vehicle – this may be:
 - In a sequence determined by the driver
 - Via a nominated door/exit
- Carry other items from the vehicle – to (for example):
 - Provide shelter
 - Provide food and drink to group
 - Lighten load
- Direct passengers to a safe location (see 'Danger Zone' above) – so they:
 - Can observe proceedings
 - Are safe – in the event there is a malfunction during the recovery process
 - Are not located beneath/below the vehicle – in case it rolls or topples
 - Are not positioned in the path of intended travel of the recovery vehicle, the trapped vehicle and/or the proposed exit route.

Involving passengers in the recovery

It may be necessary or part of the tour experience to ask a passenger to assist with the recovery of a vehicle.

Always follow Tour Operator SOPs in this regard as some operators may prohibit the involvement of passengers in recoveries.

In practice, some passengers regard involvement in such activities as part of the fun and experience of their tour.

For example, passengers may be asked to:

- Act as spotters for the recovery – which may relate to:
 - Giving directions regarding wheel placement and turn of vehicle
 - Advising of take-up of slack
 - Notifying of progress using hand signals, or verbal advice including two-way radio
- Help with shovel work
- Assist with removing obstacles
- Aid in moving property to and from the vehicle.



2.3 Inspect vehicle after recovery and perform necessary maintenance and repairs before proceeding

Introduction

After recoveries it is standard practice to inspect the vehicle and, as appropriate, perform necessary maintenance and/or repairs.

This section identifies a variety of service/maintenance tasks which may be required following a vehicle recovery.



Context

It is important to understand:

- The greatest potential for maintenance and repairs follow water crossings and situations where the vehicle has been immersed in water
- The future safety and performance of the tour vehicle may depend on immediate action being taken after the recovery to 'clear and clean' the vehicle
- Tourism Operators may provide a list of checks to be undertaken – as part of their SOPs or Emergency Management planning
- Many service tasks are not undertaken immediately after the vehicle has been recovered – but may be:
 - Delayed/deferred until the vehicle returns to base
 - Undertaken only when an authorised dealer/repair agency is available.

Tasks which may be required

It is possible the following may need to be attended to after a recovery:

- Checking there is no water in the air intake
- Inspecting tyres – for damage to tread and/or sidewalls
- Re-inflating tyres which have had pressure reduced
- Removing spark plugs, or injectors and turning the engine over to remove water
- Allowing and encouraging brakes to dry out
- Testing operation of systems and instruments
- Cleaning vehicle windscreen, mirrors and windows
- Replacing plugs, or injectors
- Applying water displacement products (such as WD-40) to dry out high and low tension electrical circuits
- Re-aligning panel damage to enable free movement of wheels and steering
- Making note to have differential and transmission checked (for water) and/or serviced at nearest opportunity

- Rolling cables and ropes back onto spools of winches
- Greasing and oiling nominated parts
- Removing visible debris – from areas and parts such as:
 - Winches
 - Winch cables
 - Radiator
 - Front of vehicle
 - Engine bay
 - Wheel arches
 - Suspension/under vehicle areas.



Work Projects

It is a requirement of this Unit you complete Work Projects as advised by your Trainer. You must submit documentation, suitable evidence or other relevant proof of completion of the project to your Trainer by the agreed date.

2.1 To fulfil the requirements of this Work Project you are required to recover a 4-wheel drive vehicle approved by your Trainer/Assessor providing evidence you have:

- Recovered a vehicle from a mud bog
 - Recovered a vehicle sand trap
 - Used a winch to recover a vehicle
 - Used another vehicle to recover a vehicle
 - Used one other nominated method to recover a vehicle from a given position/set of conditions
 - Optimised passenger safety during recoveries
 - Inspected vehicles after recovery and performed necessary maintenance/repairs before proceeding.
-

Summary

Recover a 4-wheel drive vehicle

When recovering a 4-wheel drive vehicle:

- Obtain practice and experience prior to going on tour
- Maintain passenger safety at all times
- Protect property as a secondary consideration
- Inspect the conditions in which the vehicle is trapped
- Plan the recovery
- Do not rush the recovery
- Check, position and/or prepare recovery gear which is or may be needed
- Use only properly rated recovery gear
- Attach cables, ropes and straps to dedicated mounting points
- Consider lowering tyre pressure
- Use simple options before more complex ones
- Be prepared to apply a combination of recovery options
- Undertake self-recovery or use another vehicle to assist, as most appropriate
- Communicate clearly and often with all those involved
- Inspect the vehicle after recovery
- Undertake or arrange for necessary maintenance and/or repairs following a recovery.

Element 3:

Provide basic maintenance for a 4-wheel drive vehicle

3.1 Identify minor faults

Introduction

There are several ways to identify minor vehicle faults.

This section describes the standard ways to identify the presence of minor faults so they can be diagnosed.

Basic actions

There are four basic ways to identify vehicle faults:

- Talking with those who have operated/driven or been in a vehicle.

Actions required include:

- Asking open and closed questions
- Asking probing questions
- Using terminology appropriate to the individual – such as avoiding complex automotive terms if the person does not understand them
- Listening to what is said
- Getting them to point to locations of problems they have identified.

The discussion should aim to:

- Identify type and nature of fault
- Describe the fault if it cannot be identified
- Explain whereabouts it is located (if appropriate)
- Describe symptoms and indicators
- Give an idea of frequency of the problem
- Identify conditions under which fault occurs/presents
- Test driving the vehicle – to:
 - Attempt to induce the fault – which can often be a frustrating experience as a problem ‘on tour’ often fails to manifest itself on return
 - Listen to sounds made by the vehicle
 - Get a feel for the vehicle – and how it handles and responds
 - Smell any odour which may be generated by the fault
 - Duplicate causal conditions which were known to trigger the problem



- Compare the way it is 'now' to how it normally behaves – in some cases a 'fault' with a vehicle as identified by someone else can simply be a 'lack of performance' which is really not a *fault* but simply the standard operating parameters of the vehicle
- Running stationary vehicle to induce fault – which may involve:
 - Applying handbrake and/or chocking vehicle
 - Running vehicle from a cold start
 - Starting vehicle from a hot start
 - Running vehicle with nominated accessories turned 'on' (and/or 'off') as required
 - Running vehicle for nominated period of time
 - Monitoring/observing vehicle from inside and/or outside of the vehicle
- Reading reports which have been prepared on the vehicle – such as:
 - Roadworthy/unroadworthy certificates
 - Vehicle defect notices
 - Service reports
 - Daily ('Start of Day') vehicle inspection checklists
 - End of Tour reports
 - Damage reports
 - Maintenance requests
 - Usage log.



3.2 Determine cause of faults

Introduction

When a fault has been identified it next remains to determine the cause.

This section presents standard procedures used for doing so.

Context

Identifying the cause of faults may also be known as:

- ‘Diagnosing’ faults
- ‘Troubleshooting’.

In many cases the reality of this stage is:

- It is not straight-forward
- The process can be very time consuming
- It is difficult to re-produce the fault so the cause can be found
- It can be harder to determine the cause than it is to fix the fault.



SOPs for diagnosing faults

Determining the cause of vehicle faults may include:

- Analysing available information (‘evidence’) obtained from the fault identification process to provide a list of tests to perform in order to:
 - Generate/cause the fault
 - Test options regarding what might be causing the problem
 - Eliminate possible causes in a structured and sequential, one-at-a-time basis
 - Identify causes
- Using diagnostic flowcharts – as provided by manufacturers for different parts/components of the vehicle
- Using on-board diagnostics (OBD) – which require a purpose-built scanning tool to (reader) be plugged into the on-board computer to display and interpret a Diagnostic Trouble Code (DTC) which describes the fault
- Referring to service and operator manuals – as most *minor* faults can be determined with reference to these books
- Seeking additional advice from the manufacturer – which may take the form of:
 - On-line advice – using:
 - Free, password protected access to contracted assistance provided as part of new vehicle purchase
 - Fee-for-service specialist advice
 - Over the phone assistance – from the manufacturer or the agent/business that sold the vehicle

- In-person help – which may include:
 - Talking with the Service Manager at the agency
 - Taking the vehicle in to the business for one of the mechanics to look at and give free-of-charge advice on – this is a viable option where the organisation purchases significant vehicles from the one car dealership and/or where a ‘preferred supplier’ arrangement exists with the business
 - Referring to personal experience – the more experience a person has in diagnosing faults, the greater their pool of experience becomes.
- This personal experience generates a useful source of information as it is based on ‘legitimate’, first-hand practice.
- Loading vehicle onto jacks, stands or a hydraulic lift – in order to:
 - Get a better look at the vehicle and be able to see ‘up and under’ it and the engine bay. Certainly, raised vehicles always make it easier to identify the location of leaks and drips
 - Free up certain suspension components
 - Make it easier to manipulate other parts of the vehicle
 - Using applicable sensory appraisal to help determine cause – this is similar to using the senses to help identify a fault in the first place, and can (depending on the issue) rely on:
 - Sight – for noting smoke emissions, loose connections and broken and missing parts
 - Feel – for identifying vibrations
 - Smell – for detecting unusual odours from the vehicle such as running rich, ‘electrical’ and rubber smells
 - Hearing – for perceiving exhaust (‘noise emissions’) and engine problems (‘mis-fires’, ‘racing’, ‘knocks’) when the vehicle is running
 - Monitoring on-board instrument panel – which can mean watching and interpreting the readings of:
 - Temperature gauges
 - Tachometers
 - Oil pressure
 - Battery charge
 - Monitoring externally attached equipment – such as:
 - Various diagnostic tools
 - Battery/alternator testers
 - Vacuum and fuel pump testers
 - Electronic brake service tool
 - Compression test
 - Circuit testers and multi-meters.



3.3 Conduct visual inspection of and operational tests on vehicle

Introduction

A basic starting point in providing maintenance to a vehicle is to undertake a visual inspection of the vehicle and conduct basic operational tests as part of that inspection.

This section examines what may be involved in conducting a visual inspection and operational check of a vehicle prior to providing service or operating the vehicle and presents a sample vehicle inspection checklist.



Doing a visual inspection

Provision of service/maintenance to a vehicle is traditionally carried out in closed and controlled situations, even where the vehicle is on-tour.

The inspection process should/could involve:

- Looking at and under the vehicle and closely checking all aspects of the vehicle – while it is:
 - On the ground, and/or
 - On a hoist or over an inspection pit – if a hoist or inspection pit is not available the inspection can occur by using:
 - Trolley jacks
 - Car stands
 - Under-vehicle car trolleys/creepers
- Lighting the areas being inspected – with:
 - Inspection lights
 - Torches
- Using a checklist – to:
 - Provide evidence the inspection occurred
 - Note defects and matters requiring attention
- Inspecting the vehicle in pairs – so that:
 - Issues can be analysed/discussed
 - A more effective inspection takes place – that is, there is less chance something will be missed if two people do the inspection
- Looking for signs of damage to the vehicle – as these signs can indicate:
 - Where a more detailed inspection needs to occur to determine:
 - If the damage *has caused a problem*/further problem
 - Whether *the damage is a result* of another issue

- Need for immediate action – to:
 - Fix the cause
 - Take temporary action to reduce/prevent further damage
 - Keep the vehicle on-the-road/operational
 - Need to take *later* action
 - Possible cause or source of the damage
 - What may be needed/possible to prevent recurrence
- Checking for missing parts – from:
 - Inside the cabin
 - Inside the passenger area
 - On the exterior of the vehicle
 - Under the hood/bonnet
- Noting leaking fluids – as this can:
 - Identify type of fluid
 - Identify location of leak
 - Indicate severity of problem
 - Determine, to an extent, the type of problem needing to be addressed
- Identifying projections from the vehicle – this relates to:
 - Body parts which have been worked loose, damaged or snagged and are hanging loose or sticking out from beyond the normal dimensions of the vehicle presenting a danger to other vehicles, people or structures
 - Items caught up under the vehicle or under wheel wells, around wheels and brake lines, in the engine bay or from vehicle protection bars or lighting bars
- Taking immediate action to rectify problems – which can be quickly and easily fixed on-the-spot, such as:
 - Clearing blockages
 - Tightening screws, nuts, bolts and fittings
- Checking the condition/appearance of critical components of the vehicle (which may involve physical touching, moving, squeezing and removing of parts) such as:
 - Radiator hoses
 - Timing belts
 - Fan belts
 - Tyres including all spares
 - Oil, air and fuel filters
 - Battery/batteries, battery leads, posts, connections, brackets and mountings



- Checking fluid levels – of:
 - Water – radiator and battery/ies
 - Engine oil
 - Transmission fluid
 - Brake fluid
 - Power steering fluid.



Conducting basic tests

The basics should cover testing:

- Lights – in terms of:
 - Right and left indicator lights – front and rear (including trailers)
 - Headlights – left and right, low beam and high beam
 - Driving lights and fog lights – on and off; driving lights turn off when low beam selected
 - Parking lights – front and rear
 - Tail lights
 - Reversing lights
 - Brake lights
 - Number plate light – front and rear
 - Side marker lights
 - Hazard warning lights
 - Reflectors
 - Internal vehicle/cabin lights
 - Instrument lights and signals

Lights should be:

- Clean and clear
- Without cracks or damage
- Correct colour
- Firmly and correctly fitted
- Condition of vehicle pedals – for example:
 - Checking engagement point of brake and clutch pedals
 - Confirming travel of pedals and freedom from blockages/interference
 - Activating knobs and switches and confirming they function as required
- Operation of parking brake (also known as the 'hand brake') – to confirm:
 - Hand brake can be pulled on and will hold the vehicle on a nominated gradient (say, 12% gradient)
 - Hand brake can be easily and comfortably released to the full 'off' position.

Vehicle inspection checklist

A vehicle inspection checklist should be used when inspecting a vehicle to record the findings of the inspection.

The following 'Light Motor Vehicle Inspection Checklist' is taken from the 'Light Vehicle Inspection Manual' provided by the Tasmanian Department of Infrastructure, Energy and Resources, Vehicle Operations Branch available at www.transport.tas.gov.au/standards/, (pp. 12 – 13) and provides a sample checklist which may be used when inspecting a vehicle:

1.	IDENTITY	8.	UNDER BONNET CHECK	14.	EXHAUST
☐	Compliance plate	☐	Master cylinder	☐	Exhaust leaks
☐	VIN/Chassis No.	☐	Engine oil leaks	☐	Mountings
☐	Engine No.	☐	Engine fuel leaks	☐	Clearance – hoses/wiring
2.	ELECTRICAL CHECK	☐	Air cleaner	☐	Catalytic Converter
☐	Headlight/parklights	☐	Engine mountings	15.	REAR SUSPENSION
☐	Headlight alignment	☐	Exhaust system	☐	Control arms
☐	Tail lights	☐	Battery security	☐	Bushes
☐	Indicators	☐	Wiring security	☐	Ball joints
☐	Brake lights	☐	Steering shaft	☐	Springs/shock absorbers
☐	Reverse lights	☐	Steering coupling	☐	Bump stops
☐	Reflectors	☐	Power steering leaks	☐	Modifications
☐	Lens condition	9.	STEERING	☐	Suspension height __ mm
☐	Windscreen wipers	☐	Steering box	16.	WHEELS & TYRES REAR
☐	Windscreen washers	☐	Steering rack	☐	Condition
☐	Horn	☐	Linkages	☐	Tread pattern
3.	BODY CHECK	☐	Tie rod ends	☐	Tyre speed rating
☐	Rust	☐	Idler arm	☐	Tyre load rating
☐	Protrusions	10.	FRONT SUSPENSION	☐	Rims
☐	Bonnet safety catch	☐	Control arms	☐	After market tyres/rims
☐	Windows	☐	Bushes	☐	Wheel track __ mm
☐	Window tint %	☐	Ball joints	17.	FUEL TANK

4. INSIDE CHECKS	☐	Springs/shock absorbers	☐	Mounting
☐ Heater/demister	☐	Bump stops	☐	Condition
☐ Heater fan	☐	Modifications	☐	Fuel leaks
☐ ABS/SRS lights	☐	Suspension height __ mm	☐	Fuel line
☐ Pedal condition	11. WHEELS & TYRES FRONT	18. LPG GAS		
☐ Brake pedal height	☐	Condition	☐	Gas compliance certificate
☐ Handbrake operation	☐	Tread pattern	19. BODY & FRAME	
☐ Inhibitor switch (auto)	☐	Tyre speed rating	☐	Rust
☐ Steering wheel	☐	Tyre load rating	☐	Cracks
☐ Steering wheel freeplay	☐	Rims	☐	Bends/kinks
☐ Window operation	☐	After market tyres/rims	☐	Modifications
☐ Door operation	☐	Wheel track __ mm	☐	Mudguards
5. MIRRORS	12. BRAKES			
☐ Condition	☐	Pipes and hoses		
☐ Mountings	☐	Discs and drums		
6. SEATS	☐	Calliper/hose security		
☐ Condition	☐	Pads and linings		
☐ Adjustments	☐	Linkages		
☐ Mountings	13. ENGINE & DRIVELINE			
7. SEAT BELTS	☐	Mountings		
☐ Condition	☐	Oil leaks		
☐ Compliance	☐	Universal joints		
☐ Mountings	☐	Centre bearing		
☐ Locking/snatch test				

3.4 Effect necessary repairs

Introduction

From time-to-time there can be a need to undertake certain repairs to 4WDs when on tour.

This section describes a standard range of repairs identified as being historically commonplace with four-wheel drive tour vehicles.

Replacing blown globes

Globes may need to be replaced where they are damaged/blown due to age, use or as a result of travelling over rough terrain.

Always follow the vehicle manufacturer's instructions for replacing globes as access to the individual globes varies considerably.

Possible procedures to replace a light globe can be:

- Identify the light to be replaced
 - Obtain correct globe for the job – sealed units will require the entire headlamp to be replaced; semi-sealed units only require the light globe
 - Remove the light cover/dust cover
 - Unplug electrical connector
- In some cases there is a need to remove the entire light assembly to replace a light globe.
- Disconnect whatever mechanism is used to keep the old globe in place
 - Remove the old globe noting how it sits – bayonet globes are removed by rotating them anti-clockwise
 - Put the new globe in position taking care not to touch the glass part of the globe with the fingers
 - Re-attach the mechanism, electrical connection and cover
 - Test the light to make sure it is working.



See the following for more help:

<http://www.supercheapauto.com.au/how-to/Outside-the-Car/Checking-and-Changing-an-Automotive-Light-Globe.aspx?id=79> – Checking and changing an automotive light globe

<https://www.youtube.com/watch?v=6hURsa46t4M> – How to change a headlight bulb

<https://www.youtube.com/watch?v=jk1kFc56dwk> – How to replace headlight bulbs

<https://www.youtube.com/watch?v=cB99QN1AYBU> – How to install replace front lower side reflector

<https://www.youtube.com/watch?v=mgGWFqPe0Ms> – Headlight bulb replacement service: 2005 Honda CRV

<https://www.youtube.com/watch?v=5F31wzh60zo> – How to replace taillight

<https://www.youtube.com/watch?v=9POigqpLxQc> – How to replace fog light and bulb

Replacing a fan belt

Actions are:

- Release tension on old belt at bolt on tensioner rail
- Move alternator to allow free play in old belt
- Remove old belt
- Fit new belt by looping it over alternator and water pump pulleys
- Re-locate the alternator to bring tension on the new belt
- Tighten the bolt on the tensioner rail
- Run the engine to test for:
 - Squealing noise – indicating belt is slipping and needs to be re-tensioned
 - Dashboard warning lights – which may indicate a problem.



See also:

<https://www.youtube.com/watch?v=4hnOUs9eiN4> – Change a fan belt

<http://www.supercheapauto.com.au/how-to/Servicing-Mechanical/Replacing-a-Drive-Belt.aspx?id=76> – Replacing a drive belt

<https://www.youtube.com/watch?v=Wg1AX77xEBQ> – How to replace a serpentine belt

https://www.youtube.com/watch?v=01J3Fp9_bMA – How to replace a serpentine belt: Toyota Corolla

<https://www.youtube.com/watch?v=XD0LNkFYPGQ> – Toyota timing belt

<http://www.wikihow.com/Change-an-Automotive-Belt>.

Replacing a fuse

Indicative steps are:

- Locate fuse box/block
- Remove cover
- Read key on cover (or in manual) to determine the fuse to replace
- Remove fuse to be replaced
- Insert new fuse – press in position with finger
- Verify spare fuses are available for emergency use – where space provided to store them
- Test vehicle/accessory/light which was operating on this fuse.

If fuse blows immediately then this usually indicates another problem (a short) and the vehicle may need to be taken to an auto-electrician

- If light/accessory functions as required, replace fuse box/block cover.

See more at:

<https://www.youtube.com/watch?v=kOXeo-kqg68> – How to replace a blown fuse in a car

<https://www.youtube.com/watch?v=faWZmG0fXy8> – How to find and replace a blown fuse in your car or truck



Replacing coolant hoses

Steps are:

- Ensure engine has cooled before removing radiator cap
- Drain radiator – if necessary, using drain plug if fitted: there may be a requirement to retain the fluid to put back in the radiator.

Remove bottom radiator hose from radiator if there is no drain plug

- Use pliers to release clamp holding hose in place – at front and back of top and bottom hose (if changing both hoses).

It may be a good time to replace the thermostat while the top hose is off the vehicle.

- Use flat screwdriver to help loosen connection and twist hose to break seal and remove the hose
- Clean the parts to which the hose ends were secured – to remove visible debris
- Determine if new hose clamps need to be used to fit the new hose/s – use new clamps where indicated
- Position clamps (using pliers to open them up) near end of new hoses and replace hoses
- Re-attach clamps
- Fill radiator
- Run and check for leaks.

Take the time to view:

<https://www.youtube.com/watch?v=AGmQ50naiGg> – How to replace a radiator hose (upper and lower)

<https://www.youtube.com/watch?v=C64z0L9816A> – Replacing radiator hose

<https://www.youtube.com/watch?v=xH0D22ITjQU> – How to replace the upper radiator hose



Replenishing fluids

Checking and topping up water in radiator

Important points to note:

- Engines should be cold when checking and replenishing water in the radiator to avoid safety issues associated with high temperatures and water under pressure
- Check fluid levels on level ground
- Many radiators require the addition of coolant to the water when replenishing radiator levels – this may be pre-mixed, or require mixing as part of the process. Always follow instructions on the product when adding/mixing coolant
- Wear protective eyewear when handling coolant.

The process may entail:

- Removing cover/cap from engine coolant/water tank or reservoir
- Mixing coolant with water according to manufacturer's recommendations (say, 50:50) if product is not pre-mixed – the preference is to use distilled water for this purpose
- Adding mixed coolant to engine coolant/water tank or reservoir – observe 'Min – Max' marks on the tank/reservoir and do not over-fill
- Replacing and securing cover/cap.

In some situations it may be necessary (or even acceptable) to simply top-off the engine coolant/water tank or reservoir with tap water.

See a demonstration at:

<https://www.youtube.com/watch?v=vEdtHZwLKZ8> – Basic car care and maintenance: checking car radiator coolant level

Checking and topping up brake fluid

Safety gear required should include protective gloves and glasses.

Basic requirements are:

- Identify the level of the brake fluid in the brake fluid reservoir – top-up when level is below the 'Min' mark
- Clean the top of, and around, the cap of the brake fluid reservoir prior to opening the unit – to make sure no debris falls in
- Pour brake fluid direct from container into the brake fluid reservoir – take care not to spill, and use fresh product where possible
- Fill only to 'Max' line/level on the brake fluid reservoir – do not over-fill
- Replace and secure the cap of the brake fluid reservoir
- Wipe away any spills.



See practical steps at:

<https://www.youtube.com/watch?v=LZ8BZa7d0CQ> – How to check and top-up brake fluid: Nulon

<https://www.youtube.com/watch?v=nJV0J9kTZ2g> – Top up your car's brake fluid.

Checking and topping up automatic transmission fluid

Points to note are:

- Wear protective eyewear when handling transmission fluid
- Park on flat/level ground
- Make sure engine is at operating temperature when checking the fluid – not cold
- Some makes and models may need to be checked when engine is running and some do not need the engine to be running – follow requirements in Owner Manual.

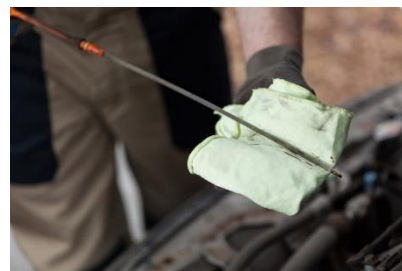
Indicative actions are:

- Locate the dipstick for the transmission fluid in the engine bay – ensure the dipstick is not the one for the engine oil
- Remove the dipstick
- Wipe clean with paper towel or clean rag
- Re-insert dipstick – to get reading
- Remove dipstick
- Read level of fluid on the dipstick – if level is below 'Min' it needs to be topped-up
- Top-up by inserting funnel into dipstick hole and pouring fluid slowly into funnel – base amount added on reading from dipstick but add small amounts
- Re-check level after each addition – avoid over-filling
- Wipe up any spills
- Replace and secure dipstick when 'Normal' level is attained.

Checking and topping up engine oil

Indicative actions are:

- Park on flat/level ground
- Check the engine oil with motor turned off and cold – do not check when engine is running and/or when it is hot
- Locate the dipstick for the engine oil in the engine bay
- Remove the dipstick
- Wipe clean with paper towel or clean rag
- Re-insert dipstick – to get reading
- Remove dipstick
- Read level of oil on the dipstick – if level is below 'Min' it needs to be topped-up



- Locate oil filler cap on top of motor – clean around the cap before removing it
- Top-up oil by inserting funnel into hole and pouring fluid slowly into funnel – base amount added on reading from dipstick
- Add small amounts each time
- Only use oil recommended by vehicle manufacturer
- Re-check level after each addition
- Do not overfill
- Clean away any spills/drips
- Replace and secure dipstick and oil filler cap when required level is attained – between 'Min' and 'Max' indicators on the dipstick.

See more at:

<https://www.youtube.com/watch?v=6IQBZm8DBAw> – Checking and adding oil to your vehicle for beginners

<http://m.supercheapauto.com.au/How-To/Oils-Lubricants/Checking-Topping-Up-Engine-Oil/194> - Checking and topping up engine oil

<https://www.youtube.com/watch?v=uvUPsk2bN0A> – Check your fluids, engine oil, transmission fluid, power steering, coolant etc.

<https://www.youtube.com/watch?v=BSlouL8Ekgw> – How to check car fluids, belts and perform full under hood inspection

Repairing punctures

If a tour vehicle suffers a puncture while on tour the standard responses are to:

- Remove the punctured tyre and replace with one of the spares carried on the vehicle
- Call in to a repair centre, tyre agency or garage as soon as possible to have the punctured tyre repaired
- Never try to repair a tyre where the sidewall is damaged.



The standard response to a puncture is, therefore, not to repair the puncture until it becomes absolutely necessary to do so.

If it is necessary the options are:

- Use a pressurised commercial can of tyre adhesive to provide a short-term solution:
 - Following the instructions on the can
 - Reducing driving speed until a professional repair can be made
 - Getting the tyre repaired professionally at the earliest opportunity

- Use a tyre repair kit to plug the hole following the instructions which come with the kit – for example:
 - Remove the cause of the puncture
 - Lubricate the reaming tool and ream out the hole with the tool in the kit – to make the hole big enough to take a the plug
 - Lubricate the plug with some rubber adhesive – and fix into the insertion tool
 - Use the insertion tool to locate the plug into the hole- - and withdraw the insertion tool
 - Cut off excess plug protruding from the tread of the tyre
 - Inflate the tyre as required.



Check the videos and articles at:

<https://www.youtube.com/watch?v=ulQmDaqv4F4> – Best tire repair video

<https://www.youtube.com/watch?v=LRQJgTxRhOA> – How to repair a puncture in a tyre with a repair plug/string

<http://www.tough toys.com.au/4wd-recovery/tyre-repair-kits-and-equipment/> - Tyre repair kits and equipment.

Bleeding brakes

If brakes become spongy, soft or ineffective on tour they may require bleeding to remove air in the brake lines.

This can be a two-person job – one person sitting in the vehicle and pumping the brake pedal and the other handling the brake bleeder bottle and the spanner to open and close the bleeder nipple/valve.

When bleeding the brakes, all four wheels have to be bled: common practice is to start the process at the wheel furthest from the master cylinder and work through all four wheels finishing with the wheel closest to the master cylinder.

As brake fluid is corrosive this means it is dangerous so appropriate protective gloves and eyewear should be used.

Indicative practice is to:

- Check the brake fluid reservoir – to make sure there is sufficient brake fluid available in the reservoir before starting the process.
Make sure the level is monitored throughout the process so air is not re-introduced into the brake system.
- Have replacement/new brake fluid on hand – to top-up the reservoir as required
- Jack the vehicle up and remove the wheels – optional in some cases
- Attach brake bleeder bottle to brake nipple
- Open valve and depress brake pedal four to five times
- Close valve and tighten – top up reservoir with brake fluid

- Check for leaks by pressing brake pedal hard – re-tighten valve/nipple if there is a leak
- Remove brake bleeder bottle – discard fluid
- Repeat for remaining three wheels – monitoring level of brake fluid in brake fluid reservoir and topping-up as necessary.

Watch the videos at:

<https://www.youtube.com/watch?v=IPwM2vn3bT0> – How to bleed disc brakes

<https://www.youtube.com/watch?v=lkd7iq-dlQ0> – Bleeding the brake system



Bleeding a diesel engine

Bleeding a diesel engine refers to removing air from the lines so the engine will re-start after, for example, the engine has run out of fuel or there has been a need to replace the fuel filter.

If the engine/fuel lines are not bled, the air in the lines blocks the flow of fuel and the engine will not start (or will run roughly).

Refer to the Owner's manual/Operator's manual for the vehicle to obtain manufacturer's instructions in this regard as there are different requirements and vehicle components for different makes and models of vehicles and engines.

More modern engines feature alternate (more simple-to-use) systems such as bleed screws whereby the screw is opened, the engine is cranked and then the screw is closed

Indicative actions may require:

- Filling the vehicle with fuel – to prevent recurrence of the problem if the problem was caused by running out of fuel
- Confirming the new fuel filter is properly fitted – and was filled with fuel prior to being re-fitted
- Prime the lift pump – by activating the lever attached to move fuel to the injector
- Open the line to the injector furthest away from the pump – and allow air to escape
- Tighten the line when there is a steady flow of diesel – and clean away spilt fuel
- Repeat for each cylinder.

Replacing spark plugs

Steps are:

- Disconnect the HT lead to the spark plug
- Use a spark plug socket attached to a wrench in an anti-clockwise direction to loosen the used spark plug – many plugs can be hard to shift so a long handle will provide the necessary extra pressure
- Use the fingers to unscrew the old spark plug and remove it once it has been freed
- Take care not to allow any debris to fall into the cylinder
- Clean the plug with a wire brush and check the condition of the plug to determine:
 - If a new plug is required
 - If there is evidence of another problem which should be investigated
- Use a feeler gauge to set the correct gap for the old or new spark plug to be fitted
- Use fingers to re-seat the plug into the block
- Tighten gently until finger tight and ensure thread is not crossed
- Finish tightening with spark plug socket attached to a torque wrench to required torque specification
- Re-attach HT lead
- Start and run motor to test.



See also:

<http://www.supercheapauto.com.au/how-to/Service-Mechanical/Replacing-Spark-Plugs.aspx?id=68> – Replacing spark plugs

<https://www.youtube.com/watch?v=jj06A6UdHvI> – Changing spark plugs and replacing plug wires.

Replacing glow plugs

Indicative actions are:

- Remove any visible debris around the plugs which may fall into the cylinders – use of compressed air is recommended
- Disconnect the (negative) battery terminal
- Use a spanner to remove the nuts holding the bridging strap to the plugs
- Remove the bridging strap
- Use a spanner to untighten the glow plug and remove with fingers
- Test the new glow plug to be fitted
- Ensure plug and thread is clean
- Screw new plug in by hand until seated in cylinder head
- Tighten with torque wrench to torque specification according to the individual plug
- Replace bridging strap securing with nuts
- Re-connect battery
- Test.

For more information see:

<https://www.youtube.com/watch?v=RCcmN-r5Z9M> – Replacing glow plugs in a Mitsubishi Delica L400

https://www.youtube.com/watch?v=4Po_1pNHE1Y – Glow plug change

https://www.youtube.com/watch?v=4Po_1pNHE1Y – Easy and complete glow plug test.

<http://autorepair.about.com/od/engine-related-diy-jobs//aa081404a.htm> - How to replace your diesel glow plugs



3.5 Effect necessary preventative maintenance

Introduction

Provision of preventative maintenance to vehicles is an ongoing necessity when on tour.

This section identifies reasons to provide preventative maintenance, explains ways to determine gives ways to determine what is required, gives generic directions about what may need to be done in this regard and addresses the need to prepare vehicles for anticipated conditions.



Need to provide preventative maintenance

Provision of regular service and maintenance to tour vehicles will/can:

- Help extend working life of the vehicle
- Increase reliability of the vehicle
- Reduce operating costs
- Present a better-looking vehicle to tour groups; this increases confidence in the tour staff and tour operator.

Ways to determine preventative maintenance requirements

The following provide a comprehensive cross-section of the options available to determine the preventative maintenance requirements for vehicles while on tour:

- Take into account distance travelled and number of hours the vehicle has been used – there is a direct relationship between extensive/long use' and the need to provide preventative maintenance
- Conduct and note response to pre-start checks – as designated by vehicle manufacturers specific to individual makes and models of 4WDs which may indicate need for maintenance. These checks commonly involving checks of:
 - Fuel, water and oil
 - Brake and transmission fluid levels
 - Battery water levels and electrolyte checks
 - Tyres
 - Belts
 - Leads
 - Hydraulic lines and connections
 - Air cleaners
 - Air-conditioners
 - Brakes

- Comply with internal organisational SOPs – many Tour Operators will provide a 'Start of day' checklist for each tour vehicle which needs to be followed before the vehicle is used every day.

These lists:

- Differ between vehicle types to reflect their individual need
- Provide a record of the inspection to prove the check was undertaken as required
- Record work done as part of the daily inspection
- Note where items may require further attention at a later stage/at conclusion of the tour

- Factor in the nature of roads/tracks travelled – some surfaces are more punishing than others and will impose a more urgent need for attention to the vehicle.

Surfaces where there have been lots of sand, rocks, water crossings and/or mud are all indicators for some extra inspection/maintenance

- Use personal observation – in terms of:
 - The way the vehicle is performing/responding
 - Sensory awareness of changes in noises, vibration and/or smells from the vehicle
 - Noticing damaged items on the vehicle
 - Noting alerts from on-board warning lights or instruments
- Refer to personal experience – related to previous incidents which have been experienced with similar vehicles under similar conditions
- Research/read manufacturer requirements in relation to routine servicing and preventative maintenance – as prescribed in:
 - Service manuals
 - Owner manuals
 - Repair manuals.



Examples of preventative maintenance which may need to be done

It is possible the following may need to be undertaken as preventative maintenance activities when on tour:

- Cleaning the vehicle – the value of cleaning a vehicle on tour must never be underestimated.

Cleaning in this context involves:

- Removing dirt, sand, mud and grit and grime
- Cleaning the engine bay
- Cleaning out under the wheel arches
- Cleaning the winch
- Cleaning lights and reflectors
- Cleaning the windscreen and windows

- Undertaking scheduled service tasks as recommended by the manufacturer – tour vehicles will generally be fully serviced prior to departing on tours but there is always the potential for ‘scheduled servicing’ to have to be provided while on tour

There may be a preference for this service to be undertaken along the route at a qualified service provider but this is not always possible.



Where it is not possible, the requirements of the Owner’s manual for the specific time period/distance travelled need to be accommodated to the best extent possible.

Indicative tasks include:

- Replacing belts
 - Replacing plugs
 - Draining and replacing fluids and oils
 - Replacing filters
- Undertaking specified additional service tasks as required by the host enterprise – these can be prescribed by the Tour Operator on the basis of previous experience with vehicles and routes.

Normally these will require:

- The same attention being paid to the same parts as for a scheduled service but with greater frequency
 - Attention to specific parts of the vehicle as a result of activities undertaken
- Replacing specific component parts – according to a nominated schedule
 - Examining the tyres – and checking and/or adjusting the tyre pressures
 - Topping up of fluids – as required based on inspection
 - Tightening loose connections and fittings
 - Replacing damaged or worn parts.

See more at:

<http://www.offroadaussie.com/2013/02/4wd-tips-and-tricks-water-crossings/> - 4WD tricks and tips: water crossing.

Preparing the vehicle

There may be a need to prepare tour vehicles in advance for known or anticipated conditions such as water and sand.

The required preparations may be:

- Dictated by Tour Operator SOPs – as developed/provided by the tour organisation
- Determined by the tour crew/vehicle driver – based on the knowledge, experience and expertise.

Refer to section 1.2 for details on preparing vehicle for water and sand.

Work Projects

It is a requirement of this Unit you complete Work Projects as advised by your Trainer. You must submit documentation, suitable evidence or other relevant proof of completion of the project to your Trainer by the agreed date.

3.1 To fulfil the requirements of this Work Project you are required to provide basic maintenance for a 4-wheel drive vehicle approved by your Trainer/Assessor providing evidence you have:

- Conducted visual inspections and/or operational tests on as vehicle to identify a nominated range of faults
- Determined the cause of a nominated range of faults
- Replaced designated parts on the vehicle
- Topped-up nominated fluids
- Changed nominated filters and plugs
- Repaired a puncture
- Replaced a nominated belt
- Replaced a nominated radiator hose
- Replaced a fuse
- Replaced a nominated globe
- Bled the brakes
- Bled a diesel engine
- Cleaned the vehicle as part of the maintenance process.

Summary

Provide basic maintenance for a 4-wheel drive vehicle

When providing basic maintenance for a 4-wheel drive vehicle:

- Use a range of options to identify vehicle faults – talk to people, drive the vehicle, run the engine, read previous reports
- Diagnose faults to determine their cause using manuals, OBD, advice from others, personal experience, observation and sensory appraisal
- Conduct regular visual inspections and checks
- Record details of inspections/checks and all work performed
- Take immediate action where necessary to prevent further damage/provide required maintenance
- Replace damaged, defective, dangerous or missing parts as necessary
- Look for signs of fluid leaks
- Conduct series of basic tests on the vehicle
- Clean the vehicle as part of regular maintenance and inspection procedures
- Prepare the vehicle ahead of known travelling conditions.

Element 4:

Report on 4-wheel drive operations

4.1 Complete documentation to record services and repair undertaken

Introduction

Standard procedure when on tour is to complete and/or retain all documentation relating to the service and other work performed on tour vehicles.

This section explains the need to maintain this paperwork, identifies documentation commonly involved, and discusses the activities necessary to comply with this requirement.



Need to maintain the paperwork

Tour Operators require this documentation to be maintained by tour crew/tour vehicle drivers in order to:

- Provide proof of inspections performed on vehicles while on tour
- Give a reference point for work which has to be performed on tour vehicles
- Allow comprehensive and accurate operational records to be maintained for each vehicle
- Enable accurate costing to be undertaken relating to the cost of operation of individual vehicles and types of vehicles
- Supply a database of information which can be provided to buyers when the tour vehicle is on-sold at the end of its useful life – to assist with proving regular maintenance and optimise re-sale value
- Help determine future repairs and maintenance which need to be provided to individual vehicles – based on the work they have done, and any service/maintenance performed on them
- Demonstrate tour crew/vehicle drivers' compliance with internal requirements – regarding daily inspections and scheduled/required maintenance and service
- Allow administration/management of the Tour Operator to reconcile receipts and invoices from external providers for vehicle expenses (see below).

Documentation folder or wallet

A wallet or folder is usually provided in all vehicles for the purposes of storing required documentation to facilitate the submission and processing of paperwork at the end of the tour.

This may contain the fuel card for the vehicle.

Keys in the process

Keys in the documentation process are:

- Write clearly
- Be honest
- Cover all work done – be comprehensive
- Use only recognised abbreviations and acronyms
- Complete required documentation 'at the time' – as opposed to relying on memory to complete it at a later time.



Daily Vehicle Inspection checklists

These are the 'Start of Day' lists to be completed for each vehicle on every day of the tour before the vehicle is used.

The checklist is used to:

- Provide a record of the inspection to prove the check was undertaken as required
- Record work done (service, repairs and maintenance) as part of the daily inspection
- Note where items may require further attention at a later stage/at conclusion of the tour.

Completing the list involves:

- A walk-around of the vehicle
- Physical inspection of the vehicle
- Starting of the engine and running the vehicle
- Monitoring gauges/instruments
- Checking leads and connections
- Testing the operation of certain systems and/or items of equipment.

Vehicle Logs

The Vehicle Log/Vehicle Usage Log records basic travel information for each vehicle on a day-by-day basis.

It commonly has to be filled in:

- At the start of the day
- At the end of the day.

Information required may vary between operators but traditionally includes:

- Name/title of tour
- Details of vehicle – usually by registration number, make and model
- Odometer reading/s – start and end of day
- Distance travelled – calculated from odometer readings
- Date/s

- Destinations and/or route taken – listing towns/locations: some Tour Operators may require odometer readings between nominated locations/destinations
- Name of driver/s – and signature
- Time/s – for travel (start and finish).

Receipts and invoices

Where money is spent on the tour vehicle during the tour for service the documentation supplied by the third party provider must be maintained for later reconciliation.

These documents (may) include quotations, receipts, service reports and invoices.

They can relate to:

- Work performed on the vehicle while on tour – such as repairs, service and maintenance
- Vehicle-related purchases/consumables – such as fuel, oil, tyres.



4.2 Notify management of issues that require external attention

Introduction

Drivers of all tour vehicles have responsibility for communicating vehicle issues to management.

This section discusses the why, when and how of this requirement.

Reasons to report

It is important to advise management of all vehicle issues which require future attention by an external service provider so as they can:

- Decide if they want to have the matter attended to or not – in some cases they may make an intentional decision *not* to fix an identified problem
- Obtain quotations – for work which has to be done so the best/most competitive price is obtained
- Make sure the work to be done is best/most efficiently integrated into the known tour schedule – of the business
- Make arrangements for a replacement vehicle – where necessary
- Ensure all tour vehicles are maintained in a safe and reliable condition – to discharge Duty of Care responsibilities and avoid preventable break-downs and delays
- Factor this knowledge into other tour-related planning decisions – such as, for example:
 - Decisions about whether or not to increase tour price based on increased vehicle-related costs
 - Decisions about whether or not to change routes and/or destinations because of damage caused to vehicles
 - Provision of increased or different training for drivers – such as driver training or mechanical training.



Timing and methods of notification

The organisational protocols (policies and procedures) of the individual Tour Operator will determine the exact requirements regarding timing of notifications.

On-tour

Usually the following applies when on-tour:

- Major repairs (which are usually defined in monetary terms, such as 'work in excess of X value) – must be notified **prior to getting work done** so that:
 - Authorisation to proceed can be obtained – because the Tour Operator may elect not to proceed with repairs
 - Arrangements for payment can be made – between the Tour Operator and the service provider

- Significant other work/expenses which have been incurred need to be communicated on the day they were incurred – as part of the standard *daily* check in/tour report that the Tour Leader may be required to make, so the Tour Operator is kept informed of issues and progress.

This report may be made:

- Via two-way radio
- Using cell phone
- Via a mobile device.

At completion of tour

When the tour concludes there can be a need for the vehicle driver to:

- Provide a comprehensive verbal report on relevant issues – as part of the Tour de-briefing/end-of-tour report, supplemented by and with reference to relevant documentation (if any) from service providers gathered throughout the tour
- Complete a vehicle maintenance requisition – specifying:
 - Identification of vehicle to be maintained
 - Nature of the problem
 - Cause of the problem, if known
 - Work already done
 - Suggestions for maintenance made by professional third party providers *en route*
- Discussing the problem with specialists – which is best done:
 - Face-to-face
 - While the vehicle is present – so specific points/parts can be indicated and problems replicated in front of mechanics/service providers.



Work Projects

It is a requirement of this Unit you complete Work Projects as advised by your Trainer. You must submit documentation, suitable evidence or other relevant proof of completion of the project to your Trainer by the agreed date.

4.1 To fulfil the requirements of this Work Project you are required to complete documentation related to reporting on 4-wheel drive operations providing evidence you have:

- Completed a vehicle usage log for a real/simulated tour
 - Completed a daily vehicle inspection checklist for a nominated 4-wheel drive vehicle
 - Completed a vehicle maintenance requisition
 - Provided a real/simulated verbal daily vehicle report
 - Provided a real/simulated verbal end-of-tour vehicle report.
-

Summary

Report on 4-wheel drive operations

When reporting on 4-wheel drive operations:

- Follow organisational protocols
- Complete all required documentation
- Include all required detail
- Communicate clearly and honestly
- Be comprehensive
- Adhere to timing requirements for completion of all documents
- Retain and submit all receipts and invoices
- Forward completed documentation as required for processing
- Notify management of vehicle issues requiring further and/or external attention
- Make verbal reports as and when required supplemented by supporting documentation.

Presentation of written work

1. Introduction

It is important for students to present carefully prepared written work. Written presentation in industry must be professional in appearance and accurate in content. If students develop good writing skills whilst studying, they are able to easily transfer those skills to the workplace.

2. Style



Students should write in a style that is simple and concise. Short sentences and paragraphs are easier to read and understand. It helps to write a plan and at least one draft of the written work so that the final product will be well organised. The points presented will then follow a logical sequence and be relevant. Students should frequently refer to the question asked, to keep 'on track'. Teachers recognise and are critical of work that does not answer the question, or is 'padded' with irrelevant material. In summary, remember to:

- Plan ahead
- Be clear and concise
- Answer the question
- Proofread the final draft.

3. Presenting Written Work

Types of written work

Students may be asked to write:

- Short and long reports
- Essays
- Records of interviews
- Questionnaires
- Business letters
- Resumes.



Format

All written work should be presented on A4 paper, single-sided with a left-hand margin. If work is word-processed, one-and-a-half or double spacing should be used. Handwritten work must be legible and should also be well spaced to allow for ease of reading. New paragraphs should not be indented but should be separated by a space. Pages must be numbered. If headings are also to be numbered, students should use a logical and sequential system of numbering.

Cover Sheet

All written work should be submitted with a cover sheet stapled to the front that contains:

- The student's name and student number
- The name of the class/unit
- The due date of the work
- The title of the work
- The teacher's name
- A signed declaration that the work does not involve plagiarism.

Keeping a Copy

Students must keep a copy of the written work in case it is lost. This rarely happens but it can be disastrous if a copy has not been kept.

Inclusive language

This means language that includes every section of the population. For instance, if a student were to write 'A nurse is responsible for the patients in her care at all times' it would be implying that all nurses are female and would be excluding male nurses.

Examples of appropriate language are shown on the right:

Mankind	<i>Humankind</i>
Barman/maid	<i>Bar attendant</i>
Host/hostess	<i>Host</i>
Waiter/waitress	<i>Waiter or waiting staff</i>

Recommended reading

Note: all Recommended Reading is sourced from 'Trove: National Library of Australia' at <http://trove.nla.gov.au/>.

Duffy, James E 2015, *Auto engine repair*, 6th edition, Tinley Park, Illinois Goodheart-Willcox Company, Inc.

Duffy, James E 2015, *Auto electricity and electronics*, Sixth Edition, Tinley Park, Illinois Goodheart-Willcox Company

ERJAVEC, JACK 2015, *Manual transmissions and transaxles: Classroom manual + shop manual*, Cengage learning, [s.l.]

GILLES, TIM 2014, *Automotive engines: Diagnosis, repair, rebuilding*, Delmar Cengage learning, [s.l.]

HADFIELD, CHRIS 2016, *Basic automotive service & systems*, Cengage learning, Boston

Halderman, James D 2015, *Automotive engines: theory and servicing*, Eighth edition, Boston Pearson

Halderman, James D 2016, *Automotive technology: principles, diagnosis, and service*, Fifth edition, Boston Pearson

HALDERMAN, JAMES D 2014, *Automotive steering, suspension & alignment*, Prentice Hall, Upper Saddle River

Seymour, Brendan, (editor.) & Rooth, John, (editor.) & Cahill, Graham, (editor.) 2014, *Roothy's ultimate 4WDing guide*, EMG Express Media Group, Silverwater, N.S.W

Stockel, Martin W & Stockel, Martin T & Johanson, Chris 2015, *Auto fundamentals : how and why of the design, construction, and operation of automobiles : applicable to all makes and models*, 11th edition, Tinley Park, IL The Goodheart-Willcox Company, Inc

Trainee evaluation sheet

Operate and maintain a 4-WD vehicle in safe working condition

The following statements are about the competency you have just completed.

Please tick the appropriate box	Agree	Don't Know	Do Not Agree	Does Not Apply
There was too much in this competency to cover without rushing.	☐	☐	☐	☐
Most of the competency seemed relevant to me.	☐	☐	☐	☐
The competency was at the right level for me.	☐	☐	☐	☐
I got enough help from my trainer.	☐	☐	☐	☐
The amount of activities was sufficient.	☐	☐	☐	☐
The competency allowed me to use my own initiative.	☐	☐	☐	☐
My training was well-organised.	☐	☐	☐	☐
My trainer had time to answer my questions.	☐	☐	☐	☐
I understood how I was going to be assessed.	☐	☐	☐	☐
I was given enough time to practice.	☐	☐	☐	☐
My trainer feedback was useful.	☐	☐	☐	☐
Enough equipment was available and it worked well.	☐	☐	☐	☐
The activities were too hard for me.	☐	☐	☐	☐

The best things about this unit were:

The worst things about this unit were:

The things you should change in this unit are:



Trainee self-assessment checklist

As an indicator to your Trainer/Assessor of your readiness for assessment in this unit please complete the following and hand to your Trainer/Assessor.

Operate and maintain a 4-WD vehicle in safe working condition

		Yes	No*
Element 1: Operate a 4-wheel drive vehicle			
1.1	Engage 4-wheel drive correctly and when appropriate	☐	☐
1.2	Drive 4-wheel drive vehicle in a variety of on-road conditions and off-road conditions	☐	☐
1.3	Provide safety and explanatory advice to passengers	☐	☐
1.4	Ensure minimal environmental impact when driving a 4-wheel drive vehicle	☐	☐
1.5	Comply with road safety and road use rules and regulations	☐	☐
Element 2: Recover a 4-wheel drive vehicle			
2.1	Use a variety of techniques to recover 4-wheel drive vehicle from bogged or trapped situations	☐	☐
2.2	Optimise safety for passengers and protection of property	☐	☐
2.3	Inspect vehicle after recovery and perform necessary maintenance and repairs before proceeding	☐	☐
Element 3: Provide basic maintenance for a 4-wheel drive vehicle			
3.1	Identify minor faults	☐	☐
3.2	Determine cause of faults	☐	☐
3.3	Conduct visual inspection of and operational tests on vehicle	☐	☐
3.4	Effect necessary repairs	☐	☐
3.5	Effect necessary preventative maintenance	☐	☐
Element 4: Report on 4-wheel drive operations			
4.1	Complete documentation to record services and repair undertaken	☐	☐
4.2	Notify management of issues that require external attention	☐	☐

Statement by Trainee:

I believe I am ready to be assessed on the following as indicated above:

Signed: _____

Date: _____

Note:

For all boxes where a **No*** is ticked, please provide details of the extra steps or work you need to do to become ready for assessment.

